Numerical Methods for PDEs

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1 Well-posed problems and reliable discretizations

1.1 Why numerical methods?

Analytical theory answers structural questions: Does a solution exist? Is it unique? How does it depend on the data? Even when those questions have satisfactory answers, a closed formula is rarely available. Moreover, the solution belongs to an infinite-dimensional function space. A numerical method replaces the continuous problem by a finite-dimensional problem whose solution can be computed.

Write an abstract problem as

$$\mathcal{P}(u, g) = 0, \tag{1.1}$$

where g collects the forcing, coefficients, domain, and boundary data. When the solution is unique, the solution operator is denoted by $u = R(g)$.

Definition 1.1 (Hadamard well-posedness)**.** Problem (1.1) is well posed in specified data and solution norms if:

1. a solution exists for every admissible datum;
2. the solution is unique;
3. the solution depends continuously on the data.

In a locally Lipschitz setting, continuous dependence takes the form

$$\|R(g + \delta g) - R(g)\|_X \leq C_g \|\delta g\|_Y$$

for all sufficiently small admissible perturbations δg. For nonzero g and $R(g)$, a local relative condition number is

$$\kappa_R(g) = \limsup_{\varepsilon \downarrow 0} \sup_{0 < \|\delta g\|_Y \leq \varepsilon \|g\|_Y} \frac{\|R(g + \delta g) - R(g)\|_X \,/\, \|R(g)\|_X}{\|\delta g\|_Y \,/\, \|g\|_Y}.$$

A large condition number signals intrinsic sensitivity. *No numerical algorithm can remove sensitivity that already belongs to the continuous problem.*

1.2 Discrete problems

Let $h > 0$ denote a resolution parameter, usually a mesh width. For each h, let X_h be the finite-dimensional space of discrete unknowns and let Y_h be the space in which the discrete equations, residuals, and right-hand sides are measured. Both spaces are equipped with suitable norms. Once the discrete data g_h are fixed, the discrete operator acts as

$$\mathcal{P}_h(\,\cdot\,, g_h) : X_h \longrightarrow Y_h.$$

The numerical problem is

$$\mathcal{P}_h(u_h, g_h) = 0, \tag{1.2}$$

with $u_h \in X_h$. We first use the simplest notation, in which the exact solution can be inserted into the discrete equations and compared directly with u_h. The common situation in which sampling or interpolation is required is explained just after the basic error argument.

Definition 1.2 (Consistency)**.** Let u be a sufficiently smooth exact solution, and let g_h denote the discrete representation of the exact data. The residual obtained by substituting u into the scheme is the *truncation error*

$$\tau_h = \mathcal{P}_h(u, g_h) \in Y_h.$$

The method is consistent if

$$\|\tau_h\|_{Y_h} \longrightarrow 0 \qquad \text{as } h \to 0.$$

Definition 1.3 (Stability)**.** A family of discrete problems is stable if perturbations of the discrete data produce uniformly controlled perturbations of the discrete solution. For a linear operator $A_h : X_h \to Y_h$ and the problem $A_h u_h = f_h$, a standard form is

$$\left\|A_h^{-1}\right\|_{\mathcal{L}(Y_h, X_h)} \leq C \qquad \text{with } C \text{ independent of } h.$$

Definition 1.4 (Convergence)**.** A method is convergent if the numerical solution approaches the exact solution in the chosen solution norm:

$$\|u_h - u\|_{X_h} \longrightarrow 0 \qquad \text{as } h \to 0.$$

Proposition 1.5 (The basic error mechanism for linear problems)**.** *Suppose $A_h : X_h \to Y_h$, $A_h u_h = f_h$, and the exact solution can be inserted into the scheme so that $A_h u = f_h + \tau_h$. If $\left\|A_h^{-1}\right\| \leq C$ uniformly, then*

$$\|u - u_h\|_{X_h} \leq C \|\tau_h\|_{Y_h}.$$

Thus consistency plus uniform stability implies convergence.

Proof. The error $e_h = u - u_h$ satisfies $A_h e_h = \tau_h$, so $e_h = A_h^{-1}\tau_h$. □

Sampling, interpolation, and the approximation property

The preceding notation deliberately suppresses a technical point. In many methods, u is a function whereas an element of X_h is a vector of nodal values, cell averages, or coefficients. The expression $A_h u$ and the difference $u_h - u$ are then not literally defined.
Introduce a sampling or interpolation operator

$$I_h : X \longrightarrow X_h.$$

The precise versions of consistency and discrete convergence are then

$$\|A_h I_h u - f_h\|_{Y_h} \longrightarrow 0, \qquad \|u_h - I_h u\|_{X_h} \longrightarrow 0.$$

(Using the PDE point-wise gives $f_h = I_h A u$. Thus, consistency is a statement about the difference $A_h I_h u - I_h A u$.) Stability gives the second statement from the first exactly as above. To conclude that the numerical method approximates the function u, one also requires an *approximation property*. If $X_h \subset X$, a typical assumption is

$$\|u - I_h u\|_X \longrightarrow 0.$$

With compatible norms, the triangle inequality then gives

$$\|u - u_h\|_X \leq \|u - I_h u\|_X + \|I_h u - u_h\|_X \longrightarrow 0.$$

If X_h consists only of grid vectors, use a reconstruction $R_h : X_h \to X$. The corresponding approximation property is $\|u - R_h I_h u\|_X \to 0$; if R_h is uniformly bounded, then

$$\|u - R_h u_h\|_X \leq \|u - R_h I_h u\|_X + C_R \|I_h u - u_h\|_{X_h} \longrightarrow 0.$$

Thus, in the fully precise formulation, approximation, consistency, and stability together imply convergence to the continuous solution.

Scope of the Lax equivalence theorem

The slogan *consistency plus stability implies convergence* is often associated with the Lax–Richtmyer theorem. That theorem is classically formulated for linear, well-posed initial-value problems and should not be quoted as an unrestricted theorem for every PDE discretization. This is in contrast with the Dahlquist Equivalence Theorem on numerical methods for ODEs.

Exercises

Exercise 1.1 (Absolute versus relative conditioning)**.** Let $R(g) = g^{-1}$ for $g > 0$. Compute the absolute sensitivity $|R'(g)|$ and the relative condition number. Explain why the two quantities tell different stories near $g = 0$.

Exercise 1.2 (A scalar model)**.** Consider $a_h u_h = f_h$ with $a_h > 0$, $a_h \to a > 0$, and $f_h \to f$. Identify a consistency error and a stability constant, and prove $u_h \to f/a$.

Exercise 1.3 (Residual versus error)**.** Construct a 2×2 linear system for which a small residual does not imply a small error in the Euclidean norm. Relate the amplification to the condition number of the matrix.

Exercise 1.4 (Nonlinear extension)**.** Suppose $F_h : X_h \to X_h$ is differentiable near a discrete solution and $F_h'(u_h)^{-1}$ is uniformly bounded. Use a first-order Taylor expansion to formulate a local analogue of the basic error mechanism.

2 Elliptic tools: Green identities and maximum principles

2.1 Gauss–Green formulas

Let $\Omega \subset \mathbb{R}^d$ be bounded with sufficiently regular boundary, and let n denote the outward unit normal. For $u \in C^1(\overline{\Omega})$, the divergence theorem gives

$$\int_\Omega \partial_i u \, \mathrm{d}x = \int_{\partial\Omega} u \, n_i \, \mathrm{d}s, \qquad i = 1, \dots, d. \tag{2.1}$$

Consequently,

$$\int_\Omega \nabla u \, \mathrm{d}x = \int_{\partial\Omega} u \, n \, \mathrm{d}s, \tag{2.2}$$

$$\int_\Omega \Delta u \, \mathrm{d}x = \int_{\partial\Omega} \partial_n u \, \mathrm{d}s, \tag{2.3}$$

$$\int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}x = -\int_\Omega v \, \Delta u \, \mathrm{d}x + \int_{\partial\Omega} v \, \partial_n u \, \mathrm{d}s. \tag{2.4}$$

The last identity is Green's first identity and is the bridge from strong to weak formulations.

2.2 Harmonic functions

A function $u \in C^2(\Omega)$ is harmonic if $\Delta u = 0$ in Ω. Harmonic functions satisfy the mean-value property:

$$u(x) = \fint_{\partial B_r(x)} u \, \mathrm{d}s = \fint_{B_r(x)} u \, \mathrm{d}y \qquad \text{whenever } \overline{B_r(x)} \subset \Omega.$$

Conversely, a continuous function with the mean-value property is harmonic.

Theorem 2.1 (Maximum principles for harmonic functions)**.** *Let Ω be connected and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ be harmonic. Then*

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u, \qquad \min_{\overline{\Omega}} u = \min_{\partial\Omega} u.$$

If an extremum is attained at an interior point, then u is constant.

The strong statement follows from the mean-value property: if $u(x_0)$ equals the global maximum, then the average of u over every small ball about x_0 can equal that maximum only when u is constant on the ball. Connectedness propagates the conclusion.

Corollary 2.2 (Positivity)**.** *If u is harmonic, $u = g \geq 0$ on $\partial\Omega$, and g is not identically zero, then $u > 0$ in Ω.*

2.3 Poisson's equation: uniqueness and stability

Consider

$$-\Delta u = f \quad \text{in } \Omega, \qquad u = g \quad \text{on } \partial\Omega. \tag{2.5}$$

Corollary 2.3 (Uniqueness)**.** *There is at most one classical solution of (2.5).*

Proof. The difference of two solutions is harmonic and vanishes on $\partial\Omega$; the maximum principle applied to w and $-w$ gives $w = 0$. □

An equivalent energy proof uses (2.4):

$$0 = -\int_\Omega w\Delta w \, \mathrm{d}x = \int_\Omega |\nabla w|^2 \, \mathrm{d}x,$$

because $w = 0$ on the boundary.

Theorem 2.4 (A maximum-norm a priori estimate)**.** *Suppose Ω is contained in a ball $B_R(x_0)$ and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ solves (2.5). Then*

$$\|u\|_{L^\infty(\Omega)} \le \|g\|_{L^\infty(\partial\Omega)} + \frac{R^2}{2d} \|f\|_{L^\infty(\Omega)}.$$

Proof. Use the barrier $\phi(x) = (R^2 - |x - x_0|^2)/(2d)$, for which $-\Delta\phi = 1$ and $\phi \ge 0$ on Ω, and apply the maximum principle to suitable combinations of u, ϕ, and the boundary norm of g. □

If $u, \widetilde{u}$ correspond to data (f, g) and $(\widetilde{f}, \widetilde{g})$, the same estimate gives

$$\|u - \widetilde{u}\|_{L^\infty(\Omega)} \le \|g - \widetilde{g}\|_{L^\infty(\partial\Omega)} + \frac{R^2}{2d} \left\|f - \widetilde{f}\right\|_{L^\infty(\Omega)}.$$

This is continuous dependence, hence the stability part of well-posedness.

2.4 A general second-order operator

For clarity, write a linear operator in nondivergence form. Let $\Omega \subset \mathbb{R}^d$ be bounded and let

$$\mathcal{L}u = -\sum_{i,j=1}^{d} a_{ij}(x)\partial_{ij}u + \sum_{i=1}^{d} b_i(x)\partial_i u + c(x)u.$$

Assume that $A = (a_{ij})$ is symmetric and uniformly elliptic:

$$\lambda |\xi|^2 \le \xi^\top A(x)\xi \le \Lambda |\xi|^2, \qquad \lambda > 0.$$

Theorem 2.5 (Weak maximum and comparison principles)**.** *If $a_{ij}, b_i, c \in C(\overline{\Omega})$, with $c \ge 0$, then, for every $u \in C^2(\Omega) \cap C(\overline{\Omega})$:*

1. *if $\mathcal{L}u \le 0$ in Ω, then*

$$\max_{\overline{\Omega}} u \le \max\left\{0, \max_{\partial\Omega} u\right\};$$

   if $c \equiv 0$, the zero may be omitted;

2. *if $\mathcal{L}u \le \mathcal{L}v$ in Ω and $u \le v$ on $\partial\Omega$, then $u \le v$ in $\overline{\Omega}$.*

Remark 2.6. Analogous results hold for divergence-form operators $-\nabla \cdot (A\nabla u) + b \cdot \nabla u + cu$, but the natural weak hypotheses and proof are formulated in Sobolev spaces.

Exercises

Exercise 2.1 (Green's second identity)**.** Derive

$$\int_\Omega (u\Delta v - v\Delta u) \, \mathrm{d}x = \int_{\partial\Omega} (u\partial_n v - v\partial_n u) \, \mathrm{d}s.$$

Exercise 2.2 (Maximum-principle uniqueness)**.** Let $c \geq 0$. Prove uniqueness for $\mathcal{L}u = f$ with prescribed Dirichlet data by applying the comparison principle to the difference of two solutions.

Exercise 2.3 (Constructing a barrier)**.** For $\Omega = (0,1)^d$, construct a quadratic $\phi \geq 0$ satisfying $-\Delta\phi = 1$. Compare the resulting stability constant with the ball-based constant above.

Exercise 2.4 (Why $c \geq 0$?)**.** Give a one-dimensional example showing that the maximum principle can fail when the zeroth-order coefficient is sufficiently negative.

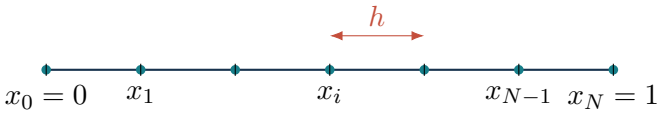
3 Centered finite differences in one dimension

3.1 The model problem and its discretization

Consider

$$-u'' = f \quad \text{in } (0,1), \qquad u(0) = u(1) = 0. \tag{3.1}$$

Choose $N \in \mathbb{N}$, set $h = 1/N$, and define $x_i = ih$, $i = 0, \dots, N$.



The centered second difference is

$$\delta_h^2 v(x) = \frac{v(x+h) - 2v(x) + v(x-h)}{h^2}.$$

The finite-difference scheme seeks nodal values $U_i \approx u(x_i)$ such that

$$(\mathcal{L}_h U)_i := -\delta_h^2 U_i = f_i := f(x_i), \quad i = 1, \dots, N-1, \qquad U_0 = U_N = 0. \tag{3.2}$$

In matrix form,

$$A_h \mathbf{U} = \mathbf{F}, \qquad A_h = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

The matrix is symmetric positive definite.

3.2 The discrete maximum principle

Theorem 3.1 (Discrete maximum principle)**.** *Let $V = (V_0, \dots, V_N)$. If $(\mathcal{L}_h V)_i \le 0$ for $i = 1, \dots, N-1$, then*

$$\max_{0 \le i \le N} V_i = \max\{V_0, V_N\}.$$

If an interior maximum is attained and $\mathcal{L}_h V \le 0$, then the maximum propagates through neighboring nodes; hence V is constant.

Proof. At an interior maximizer n,

$$0 \le -h^2 (\mathcal{L}_h V)_n = V_{n+1} - 2V_n + V_{n-1},$$

whereas maximality gives $(V_{n+1} + V_{n-1})/2 \le V_n$. Equality is forced, so both neighbors also attain the maximum. Repetition reaches the boundary. □

3.3 A sharp stability estimate

Set $w(x) = x(1-x)$ and $W_i = w(x_i)$. Since w is quadratic,

$$\mathcal{L}_h W_i = -w''(x_i) = 2, \qquad W_0 = W_N = 0, \qquad 0 \le W_i \le \frac{1}{4}.$$

For an arbitrary grid function V, apply the discrete maximum principle to

$$Z_i^{\pm} = \pm V_i - \frac{1}{2}\|\mathcal{L}_h V\|_{\ell^\infty} W_i.$$

This yields:

Lemma 3.2 (Maximum-norm stability)**.** *For every grid function V,*

$$\|V\|_{\ell^\infty} \le \max\{|V_0|, |V_N|\} + \frac{1}{8}\|\mathcal{L}_h V\|_{\ell^\infty}. \tag{3.3}$$

In particular, the homogeneous discrete problem has only the zero solution. Since the system is square and finite dimensional, uniqueness also implies existence.

3.4 Consistency and convergence

For $u \in C^4([0,1])$, Taylor's theorem with remainder gives

$$u''(x) - \delta_h^2 u(x) = -\frac{h^2}{24}\left(u^{(4)}(\xi_-) + u^{(4)}(\xi_+)\right), \tag{3.4}$$

where $\xi_- \in (x-h, x)$ and $\xi_+ \in (x, x+h)$. Therefore

$$\left|u''(x) - \delta_h^2 u(x)\right| \le \frac{h^2}{12}\left\|u^{(4)}\right\|_{L^\infty(0,1)}.$$

Note that the remainder in (3.4) carries a minus sign when written as $u'' - \delta_h^2 u$. The sign does not affect the norm bound, but it matters in a pointwise asymptotic expansion.

Let $e_i = u(x_i) - U_i$. Since $e_0 = e_N = 0$,

$$(\mathcal{L}_h e)_i = (\mathcal{L}_h I_h u)_i - f_i = u''(x_i) - \delta_h^2 u(x_i).$$

Combining the truncation bound with (3.3) proves:

Theorem 3.3 (Second-order maximum-norm convergence)**.** *If $u \in C^4([0,1])$ solves (3.1) and U solves (3.2), then*

$$\max_{0 \le i \le N} |u(x_i) - U_i| \le \frac{h^2}{96}\left\|u^{(4)}\right\|_{L^\infty(0,1)}. \tag{3.5}$$

The reusable proof pattern

error equation + uniform stability
+ truncation estimate $\Longrightarrow$ convergence estimate.

3.5 Conditioning and solver cost

Spectral estimate for the Poisson matrix

The eigenvalues of A_h are

$$\lambda_k = \frac{4}{h^2}\sin^2\left(\frac{k\pi}{2N}\right), \qquad k = 1, \dots, N-1.$$

Hence

$$\operatorname{cond}_2(A_h) \sim \frac{4}{\pi^2 h^2}.$$

Refinement improves approximation but worsens algebraic conditioning. This observation motivates preconditioners and multigrid methods.

Exercises

Exercise 3.1 (Exactness on quadratics)**.** Prove that $\delta_h^2 p = p''$ for every polynomial of degree at most three. Find the error explicitly for $p(x) = x^4$.

Exercise 3.2 (A nonzero boundary condition)**.** Derive the matrix system for $u(0) = \alpha$, $u(1) = \beta$. Show how the boundary values enter only the first and last entries of the right-hand side.

Exercise 3.3 (Energy stability)**.** Multiply $(\mathcal{L}_h U)_i = f_i$ by hU_i, sum over the interior nodes, and prove a discrete integration-by-parts identity. Deduce a stability estimate in the discrete H^1 seminorm.

Exercise 3.4 (Observed order)**.** Use the exact solution $u(x) = \sin(\pi x)$. Compute the discrete solution for successively halved h, and verify that $\|I_h u - U\|_{\ell^\infty}$ decreases by a factor close to four.

4 Weak formulations and energy principles

4.1 Why the strong formulation is too restrictive

The classical problem $-u'' = f$ appears to require two derivatives of u and pointwise values of f. Many physically natural loads do not produce such smooth solutions.

Example 4.1 (A discontinuous load)**.** If $f = \chi_{[0.4,0.6]}$, then the solution of $-u'' = f$, $u(0) = u(1) = 0$, is positive and piecewise quadratic. It is C^1, but its second derivative jumps at 0.4 and 0.6.

Example 4.2 (A point load)**.** Let $f = -\delta_{1/2}$. The weak solution is the negative tent function

$$u(x) = \begin{cases} -x/2, & 0 \le x \le 1/2, \\ -(1-x)/2, & 1/2 \le x \le 1. \end{cases}$$

It belongs to $H_0^1(0,1)$, but it is not classically differentiable at $x = 1/2$.

With the convention $-u'' = f$, a nonnegative transverse load produces a nonnegative displacement for homogeneous Dirichlet data. A negative tent corresponds to the negative point load $-\delta_{1/2}$, not to a positive load.

4.2 Sobolev spaces

Define

$$L^2(0,1) = \left\{ v : \|v\|_{L^2}^2 = \int_0^1 |v|^2 \, \mathrm{d}x < \infty \right\},$$

$$H^1(0,1) = \left\{ v \in L^2(0,1) : v' \in L^2(0,1) \right\}, \qquad H_0^1(0,1) = \overline{C_c^\infty(0,1)}^{H^1}.$$

In one dimension, functions in H^1 have continuous representatives, and

$$H_0^1(0,1) = \left\{ v \in H^1(0,1) : v(0) = v(1) = 0 \right\}.$$

Poincaré's inequality, cf. Theorem 9.6 below, gives

$$\|v\|_{L^2(0,1)} \le C_P \, \|v'\|_{L^2(0,1)} \qquad \forall v \in H_0^1(0,1).$$

Thus

$$\|v\|_V := \|v'\|_{L^2(0,1)}$$

is a norm on $V = H_0^1(0,1)$, equivalent to the full H^1 norm. The dual space $V' = H^{-1}(0,1)$ contains $L^2(0,1)$ and, in one dimension, point loads.

Norm versus seminorm

The quantity $\|v'\|_{L^2}$ is only a seminorm on $H^1(0,1)$, because constants have zero derivative. It becomes a genuine norm on $H_0^1(0,1)$ by Poincaré's inequality.

4.3 Derivation of the weak problem

Multiply $-u'' = f$ by a test function v satisfying $v(0) = v(1) = 0$, integrate, and use integration by parts:

$$-\int_0^1 u''v \, \mathrm{d}x = \int_0^1 u'v' \, \mathrm{d}x = \langle f, v \rangle_{V',V} .$$

The weak formulation is:

$$\text{Find } u \in V \text{ such that} \qquad a(u,v) = F(v) \quad \forall v \in V, \tag{4.1}$$

with

$$a(u,v) = \int_0^1 u'v' \, \mathrm{d}x, \qquad F(v) = \langle f, v \rangle_{V',V} .$$

Theorem 4.3 (Well-posedness of the weak problem)**.** *For every $F \in V'$, problem (4.1) has a unique solution, and*

$$\|u\|_V \le \|F\|_{V'} .$$

Proof. The bilinear form is continuous and coercive:

$$|a(w,v)| \le \|w\|_V \, \|v\|_V , \qquad a(v,v) = \|v\|_V^2 .$$

Hence, the Lax–Milgram Lemma (see later) gives existence and uniqueness. Testing with $v = u$ gives $\|u\|_V^2 = F(u) \le \|F\|_{V'} \, \|u\|_V$. □

4.4 Dirichlet's principle

Define the potential energy

$$J(v) = \frac{1}{2} a(v,v) - F(v) = \frac{1}{2} \int_0^1 |v'|^2 \, \mathrm{d}x - F(v). \tag{4.2}$$

Theorem 4.4 (Weak solution equals energy minimizer)**.** *The weak solution u of (4.1) is the unique minimizer of J over V.*

Proof. For $w \in V$,

$$J(u+w) = J(u) + a(u,w) - F(w) + \frac{1}{2} a(w,w) = J(u) + \frac{1}{2} a(w,w) \ge J(u).$$

Conversely, if u minimizes J, then the derivative of $t \mapsto J(u + tw)$ at $t = 0$ vanishes for every $w \in V$, which gives $a(u,w) = F(w)$. □

Nonhomogeneous Dirichlet data

If $u(0) = \alpha$ and $u(1) = \beta$, choose a lifting $G \in H^1(0,1)$ with the prescribed traces and write $u = G + w$, $w \in H_0^1(0,1)$. The admissible set is the affine space $G + H_0^1(0,1)$; variations still belong to $H_0^1(0,1)$.

Exercises

Exercise 4.1 (Poincaré inequality in one dimension)**.** Use $v(x) = \int_0^x v'(s)\,\mathrm{d}s$ and Cauchy–Schwarz to prove a Poincaré inequality on $H_0^1(0,1)$. Find a valid constant.

Exercise 4.2 (Point evaluation)**.** Prove that $|v(1/2)| \le C\,\|v'\|_{L^2}$ for $v \in H_0^1(0,1)$. Conclude that $\delta_{1/2} \in H^{-1}(0,1)$.

Exercise 4.3 (Energy of the tent function)**.** Verify distributionally that the tent function in the point-load example solves $-u'' = -\delta_{1/2}$. Compute $J(u)$.

Exercise 4.4 (Reaction term)**.** Derive the weak and energy formulations for $-u'' + cu = f$, $c \ge 0$. Identify the coercivity constant in the H_0^1 norm used above.

5 Finite elements and finite volumes in one dimension

5.1 The conforming Galerkin idea

The finite element method restricts the weak problem to a finite-dimensional subspace $V_h \subset V$:

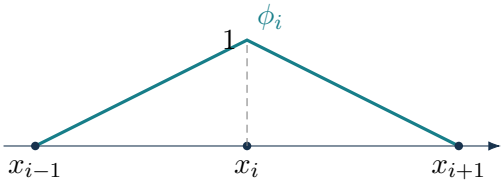
$$\text{Find } u_h \in V_h \text{ such that} \qquad a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h. \tag{5.1}$$

On the grid from section 3, choose the continuous, piecewise affine space

$$V_h = \left\{ v_h \in H_0^1(0,1) : v_h|_{[x_{i-1},x_i]} \in \mathbb{P}_1 \text{ for } i = 1, \dots, N \right\}.$$

Its nodal basis consists of hat functions ϕ_i, $i = 1, \dots, N-1$, satisfying $\phi_i(x_j) = \delta_{ij}$:

$$\phi_i(x) = \begin{cases} (x - x_{i-1})/h, & x \in [x_{i-1}, x_i], \\ (x_{i+1} - x)/h, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases}$$



Writing

$$u_h = \sum_{j=1}^{N-1} U_j \phi_j$$

and testing with every ϕ_i gives

$$K\mathbf{U} = \mathbf{b}, \qquad K_{ij} = \int_0^1 \phi_j' \phi_i' \, \mathrm{d}x, \qquad b_i = F(\phi_i). \tag{5.2}$$

For a uniform grid,

$$K = \frac{1}{h} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}.$$

5.2 Well-posedness and error analysis

Because $V_h \subset V$, the same continuity and coercivity constants apply to the discrete problem. Hence (5.1) has a unique solution. Subtracting the continuous and discrete equations gives *Galerkin orthogonality*:

$$a(u - u_h, v_h) = 0 \qquad \forall v_h \in V_h. \tag{5.3}$$

Theorem 5.1 (Céa's lemma)**.** *If a is continuous with constant M and coercive with constant α, then*

$$\|u - u_h\|_V \le \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V .$$

Proof. For any $v_h \in V_h$, coercivity and (5.3) give

$$\alpha \|u - u_h\|_V^2 \le a(u - u_h, u - u_h) = a(u - u_h, u - v_h) \le M \|u - u_h\|_V \|u - v_h\|_V .$$

□

For piecewise affine elements and $u \in H^2(0,1)$, interpolation yields

$$\inf_{v_h \in V_h} \|u - v_h\|_{H^1} \le Ch \|u\|_{H^2} .$$

Thus the energy error is first order. A duality argument gives a second-order L^2 estimate under the corresponding elliptic regularity.

5.3 When finite elements reproduce finite differences

For $F(v) = \int_0^1 fv \, \mathrm{d}x$, the i-th finite element equation is

$$-\frac{1}{h}U_{i-1} + \frac{2}{h}U_i - \frac{1}{h}U_{i+1} = \int_{x_{i-1}}^{x_{i+1}} f\phi_i \, \mathrm{d}x.$$

Approximating the right-hand side by the nodal quadrature

$$\int_{x_{i-1}}^{x_{i+1}} f\phi_i \, \mathrm{d}x \approx hf(x_i)$$

and dividing by h recovers the centered finite-difference equation.

The equivalence uses quadrature

The linear finite element stiffness matrix and the centered finite-difference matrix have the same tridiagonal pattern, up to a row scaling. Exact equivalence for a general load requires a specific approximation of the finite element load vector; it is not an identity with exact integration.

5.4 A finite volume derivation

Let the control volume about x_i be $I_i = [x_{i-1/2}, x_{i+1/2}]$. Integrating $-u'' = f$ over I_i gives the exact balance

$$-u'(x_{i+1/2}) + u'(x_{i-1/2}) = \int_{I_i} f \, \mathrm{d}x. \tag{5.4}$$

Approximate the face fluxes by

$$u'(x_{i+1/2}) \approx \frac{U_{i+1} - U_i}{h}, \qquad u'(x_{i-1/2}) \approx \frac{U_i - U_{i-1}}{h},$$

and use the midpoint rule on the right. After division by h,

$$\frac{-U_{i+1} + 2U_i - U_{i-1}}{h^2} = f(x_i).$$

Thus the same algebraic scheme arises from three perspectives: pointwise differentiation, a variational principle with nodal quadrature, and local flux balance.

By construction, the finite volume approach allows to show a local conservation property as follows. Write the balance on cell i as

$$-q_{i+1/2} + q_{i-1/2} = \int_{I_i} f \, \mathrm{d}x, \qquad q_{i+1/2} \approx \frac{U_{i+1} - U_i}{h}.$$

Summing the finite volume equations over a consecutive block of control volumes, that is from $i = m$ to $i = n$, with $m < n$, gives

$$\sum_{i=m}^{n} (-q_{i+1/2} + q_{i-1/2}) = q_{m-1/2} - q_{n+1/2} = \int_{x_{m-1/2}}^{x_{n+1/2}} f \, \mathrm{d}x.$$

Every internal face flux appears once with each sign and cancels. The remaining equation says that the net outward flux through the boundary of the block equals the total source inside it. This is local conservation on any union of control volumes.

How the viewpoints differ

- Finite differences make truncation and maximum-principle arguments transparent.
- Finite elements naturally handle weak solutions, complex geometry, and energy-norm error estimates.
- Finite volumes enforce a local conservation law and are often preferred when flux balance is central.

The algebraic coincidence in this one-dimensional example should not be mistaken for complete equivalence of the methods.

Exercises

Exercise 5.1 (Element assembly)**.** Compute the 2×2 stiffness matrix on a single interval $[x_{i-1}, x_i]$, then assemble the global tridiagonal matrix.

Exercise 5.2 (Galerkin best approximation)**.** For the Poisson bilinear form, show that u_h is the orthogonal projection of u onto V_h in the energy inner product. Improve Céa's estimate to an equality involving the best approximation error.

Exercise 5.3 (A point load in FEM)**.** Take $F(v) = -v(1/2)$ on a mesh containing $1/2$. Write the finite element load vector and show that the exact tent solution belongs to V_h. What does the discrete solution equal?

6 Variable coefficients and boundary conditions in one dimension

6.1 A centered scheme

Consider the nondivergence-form boundary-value problem

$$\mathcal{L}u := -a(x)u'' + b(x)u' + c(x)u = f(x), \qquad u(0) = u_0, \quad u(1) = u_1, \tag{6.1}$$

where $a(x) \geq a_0 > 0$ and $c(x) \geq 0$. At an interior node define

$$(\mathcal{L}_h U)_i = -a_i \frac{U_{i+1} - 2U_i + U_{i-1}}{h^2} + b_i \frac{U_{i+1} - U_{i-1}}{2h} + c_i U_i. \tag{6.2}$$

The lower, diagonal, and upper coefficients in row i are

$$-\frac{a_i}{h^2} - \frac{b_i}{2h}, \qquad \frac{2a_i}{h^2} + c_i, \qquad -\frac{a_i}{h^2} + \frac{b_i}{2h}.$$

6.2 Monotonicity and the discrete maximum principle

The off-diagonal coefficients are nonpositive provided

$$a_i - \frac{h}{2}|b_i| \geq 0. \tag{6.3}$$

This is a mesh Péclet restriction: centered convection must not overwhelm diffusion at the grid scale (more in the following sections).

Theorem 6.1 (Discrete comparison principle)**.** *Suppose (6.3) holds. If $(\mathcal{L}_h V)_i \leq 0$ at all interior nodes, then*

$$\max_i V_i \leq \begin{cases} \max\{V_0, V_N\}, & c \equiv 0, \\ \max\{V_0, V_N, 0\}, & c \geq 0. \end{cases}$$

Consequently, if $\mathcal{L}_h U \leq \mathcal{L}_h W$ in the interior and $U \leq W$ on the boundary, then $U \leq W$ everywhere.

Let A_h denote the interior matrix corresponding to homogeneous Dirichlet boundary values. Then A_h has a positive diagonal and nonpositive off-diagonal entries. Its inverse is nonnegative. This is the characteristic structure of a nonsingular M-matrix[1]. The comparison principle shows that the M-matrix property is the algebraic form of monotonicity.

The discrete maximum principle first implies that A_h is nonsingular: if $A_h V = 0$, applying the principle to V and $-V$ gives $V = 0$. If $F \geq 0$ and $A_h V = F$, then $\mathcal{L}_h(-V) = -F \leq 0$, with zero boundary values. Hence $-V \leq 0$, so $V \geq 0$. Thus A_h^{-1} maps nonnegative vectors to nonnegative vectors. Applying this observation to each coordinate vector shows that $A_h^{-1} \geq 0$ componentwise. Consequently, A_h is a nonsingular M-matrix.

[1] A matrix A is an M-matrix if its off-diagonal etries satisfy $a_{ij} \leq 0$ and its inverse A^{-1} exists with $A^{-1} \geq 0$. The diagonal entries of an M-matrix are positive; that is, $a_{ii} > 0$. An equivalent condition for M-matrix (M-criterion) is the following. Assume $a_{ij} \leq 0$, $i \neq j$. Then, A is an M-matrix if and only if there exists a vector $e > 0$ such that $Ae > 0$. The barrier function used above in Section 3.3 and again below in Section 6.4 has precisely the meaning of providing such vector. IMoreover, constructing the barrier not only provides an alternative M-matrix proof but also yields a stability constant.

6.3 Consistency

For a smooth u, the centered first derivative satisfies

$$\frac{u(x+h)-u(x-h)}{2h} = u'(x) + \frac{h^2}{6}u^{(3)}(\eta)$$

for some $\eta \in (x-h, x+h)$. Combining this with (3.4) gives

$$|(\mathcal{L}_h I_h u)_i - (\mathcal{L}u)(x_i)| \le h^2 \left(\frac{\|a\|_{L^\infty}}{12} \left\|u^{(4)}\right\|_{L^\infty} + \frac{\|b\|_{L^\infty}}{6} \left\|u^{(3)}\right\|_{L^\infty} \right).$$

Note, in particular, that the reaction term $c\,u$ term is sampled exactly and thus it does not appear in the consistency estimate.

6.4 Stability

When $b = 0$, we can use again the barrier function $w = x(1-x)$ which obeys

$$(\mathcal{L}_h w)_i = 2a_i + c_i w_i \ge 2a_0.$$

The comparison argument therefore yields

$$\|V\|_{\ell^\infty} \le \max\{|V_0|, |V_N|\} + \frac{1}{8a_0} \|\mathcal{L}_h V\|_{\ell^\infty}.$$

With $b \neq 0$, things are more complicated, and a different barrier function needs to be used. Still maximum-norm stability can be obtained with a stability constant depending on a_0, b, c, and the interval, provided the monotonicity condition holds.

Theorem 6.2 (Maximum-norm stability)**.** *Assume $a_i \ge a_0 > 0$, $c_i \ge 0$, and $|b_i| \le B$, and suppose the mesh Péclet condition (6.3) holds. Then there are constants $h_0 > 0$ and $C_{\mathrm{stab}} > 0$, independent of h, such that every grid function V satisfies, for $h \le h_0$,*

$$\|V\|_{\ell^\infty} \le \max\{|V_0|, |V_N|\} + C_{\mathrm{stab}} \|\mathcal{L}_h V\|_{\ell^\infty}. \tag{6.4}$$

As before, once consistency and stability are ensured, convergence follows.

Theorem 6.3 (Second-order maximum-norm convergence)**.** *Let $u \in C^4([0,1])$ solve (6.1), and let U solve the centered scheme (6.2) with the exact Dirichlet boundary values. Under the hypotheses of the maximum-norm stability theorem, for all sufficiently small h,*

$$\|I_h u - U\|_{\ell^\infty} \le C_{\mathrm{stab}} h^2 \left(\frac{\|a\|_{L^\infty}}{12} \left\|u^{(4)}\right\|_{L^\infty} + \frac{\|b\|_{L^\infty}}{6} \left\|u^{(3)}\right\|_{L^\infty} \right). \tag{6.5}$$

In particular, the centered scheme converges with order two in the nodal maximum norm.

Proof. Set $e = I_h u - U$. The boundary values are exact, so $e_0 = e_N = 0$, and the error equation is

$$\mathcal{L}_h e = \mathcal{L}_h I_h u - I_h(\mathcal{L}u).$$

Apply (6.4) and then the preceding truncation estimate. □

Divergence form and conservative fluxes

For

$$-(a(x)u')' + c(x)u = f,$$

it is generally preferable to approximate the face flux $a\,u'$:

$$-\frac{1}{h}\left[a_{i+1/2}\frac{U_{i+1}-U_i}{h} - a_{i-1/2}\frac{U_i - U_{i-1}}{h}\right] + c_i U_i = f_i.$$

This discretization is conservative and corresponds directly to a finite volume balance (see below). It is not the same as simply sampling $-a(x)u''$.

6.5 A Neumann boundary and ghost points

Suppose the right boundary condition is $u'(1) = q$. A centered ghost relation is

$$\frac{U_{N+1} - U_{N-1}}{2h} = q.$$

For $q = 0$, substituting $U_{N+1} = U_{N-1}$ into the differential equation at x_N gives, when b_N drops out,

$$2a_N\frac{U_N - U_{N-1}}{h^2} + c_N U_N = f_N.$$

Boundary accuracy must be checked separately

The centered ghost relation is second order as an approximation to the derivative, but the eliminated boundary equation above has only first-order local truncation error in general when written in the same PDE scaling as the interior equations. Nevertheless, the ghost-point method may still be globally second-order accurate because this lower-order residual occurs only at the boundary and its influence is weakened by the discrete solution operator. Establishing second-order convergence therefore requires a boundary-sensitive stability argument, a suitably scaled boundary equation, or a discrete Green's-function estimate; it does not follow directly from the maximum norm of the local truncation errors. An alternative treatment of the Neumann condition which has a second-order truncation error can be obtained by exploiting directly the following second-order one-sided divided difference:

$$\frac{3U_N - 4U_{N-1} + U_{N-2}}{2h} = q.$$

Exercises

Exercise 6.1 (Matrix coefficients)**.** Derive the three coefficients of (6.2). Show that their row sum is c_i.

Exercise 6.2 (Mesh Péclet number)**.** For constant $a > 0$ and b, write (6.3) as a condition on $\mathrm{Pe}_h = |b|\,h/(2a)$. Numerically compare centered and first-order upwind differences when $\mathrm{Pe}_h > 1$.

Exercise 6.3 (A conservative interface problem)**.** Let a be piecewise constant with a jump at $x = 1/2$. Derive a finite volume scheme for $-(au')' = f$. Explain why the face value of a should be a harmonic rather than arithmetic average when the interface lies between nodes.

Exercise 6.4 (Neumann nullspace)**.** Consider $-u'' = f$ with $u'(0) = u'(1) = 0$. Find the compatibility condition on f, identify the continuous and discrete nullspaces, and state one normalization that restores uniqueness.

Exercise 6.5 (Second-order accuracy of the ghost-point method)**.** Consider the mixed boundary-value problem with right boundary condition $u'(1) = q$ discretised using the ghost point method. Show that the local truncation error at the right endpoint is only $O(h)$, whereas the interior truncation errors are $O(h^2)$. Prove nevertheless that

$$\max_{0 \le i \le N} |U_i - u(x_i)| \le Ch^2,$$

where C is independent of h.

7 Finite differences in several dimensions

7.1 The five-point scheme

Consider

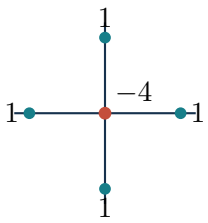
$$-\Delta u = f \quad \text{in } \Omega = (0,1)^2, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{7.1}$$

On the uniform grid $x_i = ih$, $y_j = jh$, $h = 1/N$, define

$$\Delta_h U_{ij} = \frac{U_{i+1,j} + U_{i-1,j} + U_{i,j+1} + U_{i,j-1} - 4U_{ij}}{h^2}.$$

The five-point method is

$$-\Delta_h U_{ij} = f(x_i, y_j) \quad (1 \le i, j \le N-1), \qquad U_{ij} = 0 \quad \text{on boundary nodes.} \tag{7.2}$$



With lexicographic ordering of the $(N-1)^2$ unknowns,

$$A_h = \frac{1}{h^2}(T \otimes I + I \otimes T), \qquad T = \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{pmatrix}.$$

The matrix is sparse, symmetric positive definite, and block tridiagonal.

7.2 Discrete maximum principle and stability

Theorem 7.1 (Two-dimensional discrete maximum principle)**.** *If $-\Delta_h V_{ij} \le 0$ at every interior node, then the maximum of V is attained on the grid boundary.*

Proof. At an interior maximizer,

$$V_{ij} \le \frac{1}{4}\left(V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1}\right) \le V_{ij}.$$

Equality forces all four neighbors to share the maximum. Connectivity propagates it to the boundary. □

Use the discrete barrier

$$w(x,y) = \frac{x(1-x) + y(1-y)}{4}.$$

It satisfies $-\Delta_h w = 1$, vanishes only at the four corners but is nonnegative on the whole boundary, and $\max_{\overline{\Omega}} w = 1/8$. Comparison gives

$$\|V\|_{\ell^\infty(\Omega_h)} \le \|V\|_{\ell^\infty(\partial\Omega_h)} + \frac{1}{8}\,\|-\Delta_h V\|_{\ell^\infty(\Omega_h^\circ)}\,. \tag{7.3}$$

7.3 Truncation error and convergence

Let

$$M_{4,0} = \left\|\partial_x^4 u\right\|_{L^\infty(\Omega)}, \qquad M_{0,4} = \left\|\partial_y^4 u\right\|_{L^\infty(\Omega)}.$$

Applying the one-dimensional centered-difference identity in each coordinate gives

$$\|(-\Delta_h I_h u) - f\|_{\ell^\infty} \le \frac{h^2}{12}(M_{4,0} + M_{0,4}).$$

The error equation and (7.3) yield:

Theorem 7.2 (Five-point convergence)**.** *If $u \in C^4(\overline{\Omega})$ solves (7.1), then the solution of (7.2) satisfies*

$$\max_{i,j} |u(x_i, y_j) - U_{ij}| \le \frac{h^2}{96}(M_{4,0} + M_{0,4}). \tag{7.4}$$

7.4 General elliptic operators and mixed derivatives

For a mixed derivative, the natural centered formula is

$$\partial_{xy} u(x_i, y_j) \approx \frac{U_{i+1,j+1} - U_{i+1,j-1} - U_{i-1,j+1} + U_{i-1,j-1}}{4h^2}.$$

Thus a general second-order operator produces a wider but still sparse stencil. A crucial warning is that consistency and ellipticity alone do not guarantee a discrete maximum principle. For example, discretizing

$$-a\,u_{xx} - 2b\,u_{xy} - c\,u_{yy}, \qquad a > 0, \quad ac - b^2 > 0,$$

with the formula above creates diagonal-neighbor coefficients of both signs. Monotonicity may require coefficient restrictions, a rotated stencil, or a wider stencil.

7.5 A compact fourth-order Poisson scheme

Higher order usually requires more grid points, but the Poisson equation has useful compact formulas. Define the nine-point Laplacian

$$(\Delta_9 U)_{ij} = \frac{1}{6h^2}\left[4 \sum_{\text{edge nbrs}} U + \sum_{\text{corner nbrs}} U - 20 U_{ij}\right].$$

For a smooth function,

$$\Delta_9 u = \Delta u + \frac{h^2}{12}\Delta^2 u + \mathcal{O}(h^4).$$

Since $-\Delta u = f$, a fourth-order compact discretization is

$$-\Delta_9 U = f + \frac{h^2}{12}\Delta f \quad \text{at interior nodes.} \tag{7.5}$$

The right-hand side correction is essential; applying the nine-point stencil with the unmodified right-hand side is generally only second order.

Three dimensions

The standard discretization of $-\Delta$ in three dimensions uses the center and its six coordinate neighbors. The maximum-principle argument, Kronecker-sum matrix representation, and consistency analysis extend directly, although the linear systems become much larger.

Exercises

Exercise 7.1 (Kronecker eigenpairs)**.** Use the one-dimensional eigenvectors to find every eigenvalue of the two-dimensional five-point matrix. Determine the asymptotic growth of its Euclidean condition number.

Exercise 7.2 (Rectangular grid)**.** Derive the five-point formula when $h_x \neq h_y$. Construct a quadratic barrier and obtain an explicit maximum-norm stability constant.

Exercise 7.3 (Nine-point expansion)**.** Expand all nine stencil values through sixth order and verify $\Delta_9 u = \Delta u + (h^2/12)\Delta^2 u + \mathcal{O}(h^4)$.

Exercise 7.4 (Anisotropy and monotonicity)**.** For constant a, b, c, write every coefficient in the centered stencil for $-au_{xx} - 2bu_{xy} - cu_{yy}$. Determine conditions under which all off-diagonal coefficients of the resulting matrix are nonpositive.

8 Nonuniform grids and irregular boundaries

8.1 The optimal three-point second derivative

Let

$$a = x_0 < x_1 < \cdots < x_N = b, \qquad h_i = x_i - x_{i-1}.$$

At x_i, the three-point approximation obtained by matching constants, linear functions, and quadratics is

$$\begin{aligned} D_h^2 U_i &= \frac{2}{h_i + h_{i+1}} \left(\frac{U_{i+1} - U_i}{h_{i+1}} - \frac{U_i - U_{i-1}}{h_i} \right) \\ &= \frac{2}{h_{i+1}(h_i + h_{i+1})} U_{i+1} - \frac{2}{h_i h_{i+1}} U_i + \frac{2}{h_i(h_i + h_{i+1})} U_{i-1}. \end{aligned} \tag{8.1}$$

For $-u'' = f$, set $-D_h^2 U_i = f(x_i)$. The off-diagonal coefficients of $-D_h^2$ are negative, so the discrete maximum principle survives on arbitrary positive step sizes.

8.2 What order is the nonuniform formula?

A Taylor expansion about x_i, with $h_- = h_i$ and $h_+ = h_{i+1}$, gives

$$D_h^2 u(x_i) = u''(x_i) + \frac{h_+ - h_-}{3} u^{(3)}(x_i) + \frac{h_+^2 - h_+ h_- + h_-^2}{12} u^{(4)}(x_i) + \mathcal{O}(h_{\max}^3). \tag{8.2}$$

A nonuniform three-point formula is not always second order

On an arbitrary quasi-uniform mesh, $h_+ - h_-$ can be of order $h_{\max}$, and (8.1) is then only first-order accurate locally. It is second order on a uniform mesh, or more generally on a smoothly graded mesh satisfying $|h_{i+1} - h_i| = \mathcal{O}(h_{\max}^2)$. A bounded mesh ratio alone does not cancel the cubic term.

This distinction is important: the stencil is monotone on any mesh, but its consistency order depends on how neighboring step sizes vary.

8.3 The Shortley–Weller idea

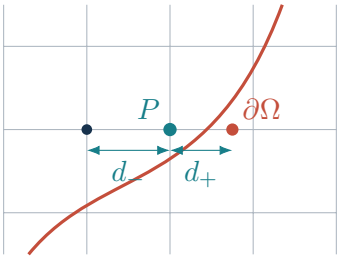
Let $\Omega \subset \mathbb{R}^2$ have a curved boundary and overlay it with a Cartesian grid. At a grid point $P \in \Omega$, a coordinate line may meet $\partial\Omega$ before reaching the next Cartesian node. Let d_- and d_+ be the distances from P to the available node or boundary intersection in the negative and positive x-directions. Use the nonuniform formula

$$D_{xx}^{SW} U(P) = \frac{2}{d_- + d_+} \left[\frac{U(P + d_+ e_x) - U(P)}{d_+} - \frac{U(P) - U(P - d_- e_x)}{d_-} \right], \tag{8.3}$$

inserting the prescribed value g when an endpoint is on the physical boundary. Define D_{yy}^{SW} analogously and solve

$$-(D_{xx}^{SW} + D_{yy}^{SW}) U(P) = f(P).$$

At regular nodes this is exactly the five-point method. At cut stencils, the local formula is generally only first order because the opposite distances are unequal.



8.4 Why first-order local error need not destroy global order

The low-order truncation error is confined to a thin set of nodes near the boundary. A generalized maximum-principle argument keeps separate track of interior and near-boundary residuals and uses different barrier functions for them. Under standard smoothness and geometric assumptions, the Shortley–Weller method can still satisfy a nodal estimate of the form

$$\|I_h u - U\|_{\ell^\infty(\Omega_h)} \le C_\Omega h^2 \|u\|_{C^4(\overline{\Omega})}.$$

The exact hypotheses matter; a local first-order statement alone neither proves nor disproves second-order global convergence.

An alternative is to use linear interpolation along a cut segment to eliminate a near-boundary unknown. Although linear interpolation sounds first order, its pointwise interpolation error for a C^2 function is $\mathcal{O}(h^2)$.

Monotonicity as a design principle

For elliptic problems, a stencil with positive diagonal, nonpositive off-diagonal coefficients, and suitable row sums often produces an M-matrix. This connects:

$$\text{discrete maximum principle} \iff \text{nonnegative inverse} \implies \text{maximum-norm stability}.$$

High order, compactness, and monotonicity can compete, especially for anisotropic operators and irregular meshes.

Exercises

Exercise 8.1 (Undetermined coefficients)**.** Starting from $\alpha U_{i+1} + \beta U_i + \gamma U_{i-1}$, impose exactness on 1, x, and x^2, and derive (8.1).

Exercise 8.2 (Local order)**.** Verify (8.2). Give one family of meshes for which the method is first order and one smoothly graded family for which it is second order.

Exercise 8.3 (Shortley–Weller coefficients)**.** Expand (8.3) into its three coefficients. Prove that the corresponding contribution to $-\Delta_h$ has nonpositive off-diagonal coefficients for every $d_-, d_+ > 0$.

Exercise 8.4 (Boundary interpolation)**.** Let x_b be a boundary intersection between two collinear points whose separation is $\mathcal{O}(h)$. Derive the remainder of linear interpolation at x_b and show that it is $\mathcal{O}(h^2)$ for $u \in C^2$.

9 Functional analytic preliminaries

Throughout, $\Omega \subset \mathbb{R}^d$ is open. It will be assumed bounded when an embedding or estimate requires $|\Omega| < \infty$, and Lipschitz when traces on $\partial\Omega$ are used. Functions that agree almost everywhere are identified.

9.1 The spaces $L^p(\Omega)$

For $1 \le p < \infty$, define

$$L^p(\Omega) = \left\{ v : \Omega \to \mathbb{R} \text{ measurable} : \int_\Omega |v|^p \, \mathrm{d}x < \infty \right\},$$

with norm

$$\|v\|_{L^p(\Omega)} = \left(\int_\Omega |v|^p \, \mathrm{d}x \right)^{1/p}.$$

The limiting space is

$$L^\infty(\Omega) = \left\{ v : \Omega \to \mathbb{R} \text{ measurable} : \operatorname*{ess\,sup}_{x \in \Omega} |v(x)| < \infty \right\},$$

with

$$\|v\|_{L^\infty(\Omega)} = \operatorname*{ess\,sup}_{x \in \Omega} |v(x)| .$$

The essential supremum, rather than the pointwise supremum, makes the norm independent of changes on sets of measure zero.

Let p' be the conjugate exponent, defined by

$$\frac{1}{p} + \frac{1}{p'} = 1,$$

with the usual conventions $1' = \infty$ and $\infty' = 1$.

Theorem 9.1 (Hölder inequality)**.** *If $v \in L^p(\Omega)$ and $w \in L^{p'}(\Omega)$, then*

$$\left| \int_\Omega vw \, \mathrm{d}x \right| \le \|v\|_{L^p(\Omega)} \, \|w\|_{L^{p'}(\Omega)} .$$

When $|\Omega| < \infty$, Hölder's inequality also allows to compares different Lebesgue spaces. If $1 \le p \le q \le \infty$, then

$$L^q(\Omega) \hookrightarrow L^p(\Omega), \qquad \|v\|_{L^p(\Omega)} \le |\Omega|^{1/p - 1/q} \, \|v\|_{L^q(\Omega)} . \tag{9.1}$$

The assumption $|\Omega| < \infty$ is essential.

9.2 The Hilbert space $L^2(\Omega)$

The space $L^2(\Omega)$ is a Hilbert space with inner product

$$(u, v)_{L^2(\Omega)} = \int_\Omega uv \, \mathrm{d}x.$$

The induced norm is the usual L^2-norm, and Hölder's inequality with $p = p' = 2$ becomes the Cauchy–Schwarz inequality

$$\left| (u, v)_{L^2(\Omega)} \right| \le \|u\|_{L^2(\Omega)} \, \|v\|_{L^2(\Omega)} .$$

Unlike continuous functions, L^2-functions may have jumps and need not possess pointwise values with any intrinsic meaning.

9.3 Dual spaces and the Riesz theorem

Let V be a normed space. Its dual space

$$V' = \mathcal{L}(V;\mathbb{R})$$

consists of the bounded linear functionals $F : V \to \mathbb{R}$, equipped with

$$\|F\|_{V'} = \sup_{v\in V\setminus\{0\}} \frac{|F(v)|}{\|v\|_V}.$$

The value $F(v)$ is also written as the duality pairing $\langle F, v\rangle_{V',V}$.

Theorem 9.2 (Riesz representation)**.** *Let H be a real Hilbert space. For every $F \in H'$, there is a unique $u_F \in H$ such that*

$$F(v) = (u_F, v)_H \qquad \forall v \in H,$$

and $\|F\|_{H'} = \|u_F\|_H$.

In particular, every functional on $L^2(\Omega)$ can be represented by a unique L^2-function:

$$F(v) = \int_\Omega f v \,\mathrm{d}x.$$

This justifies the customary identification $(L^2(\Omega))' \cong L^2(\Omega)$. For a general space V, however, elements of V' need not be ordinary functions.

9.4 Test functions and distributions

The space of test functions is

$$\mathcal{D}(\Omega) = C_c^\infty(\Omega).$$

A sequence φ_n converges to φ in $\mathcal{D}(\Omega)$ if all functions have support in one compact subset of Ω and every derivative converges uniformly there. This is the standard test-function topology.

Definition 9.3 (Distribution)**.** A distribution on Ω is a continuous linear functional $T : \mathcal{D}(\Omega) \to \mathbb{R}$. The space of distributions is denoted by $\mathcal{D}'(\Omega)$, and its action is written

$$\langle T, \varphi\rangle = T(\varphi).$$

Every $f \in L^1_{\mathrm{loc}}(\Omega)$ defines a regular distribution by

$$\langle f, \varphi\rangle = \int_\Omega f\varphi \,\mathrm{d}x.$$

Thus $L^p(\Omega) \subset \mathcal{D}'(\Omega)$ for every $1 \le p \le \infty$. Not every distribution is regular: for example, the Dirac mass at $x_0 \in \Omega$ is defined by

$$\langle \delta_{x_0}, \varphi\rangle = \varphi(x_0).$$

9.5 Distributional derivatives

For $T \in \mathcal{D}'(\Omega)$, define its distributional derivative by

$$\langle D_i T, \varphi \rangle = - \langle T, D_i \varphi \rangle \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

More generally, for a multi-index α,

$$\langle D^\alpha T, \varphi \rangle = (-1)^{|\alpha|} \langle T, D^\alpha \varphi \rangle . \tag{9.2}$$

Every distribution therefore has derivatives of every order, although those derivatives need not be representable by locally integrable functions.

Example 9.4 (The Heaviside function)**.** On $\mathbb{R}$, let $H(x) = 1$ for $x > 0$ and $H(x) = 0$ for $x < 0$. Then

$$\langle DH, \varphi \rangle = - \int_0^\infty \varphi'(x)\, \mathrm{d}x = \varphi(0) = \langle \delta_0, \varphi \rangle .$$

Hence $DH = \delta_0$ in $\mathcal{D}'(\mathbb{R})$, even though H has no classical derivative at the origin.

The above examples shows that if a function belongs to L^p, its derivatives, which are always defined in the distributional sense, may *not* belong to L^p itself.

9.6 Sobolev spaces

Definition 9.5 (Sobolev space)**.** For $k \in \mathbb{N}_0$ and $1 \le p \le \infty$, define

$$W^{k,p}(\Omega) = \{ v \in L^p(\Omega) : D^\alpha v \in L^p(\Omega) \text{ for all } |\alpha| \le k \},$$

where all derivatives are understood in the distributional sense.

When $p = 2$, write $H^k(\Omega) = W^{k,2}(\Omega)$. It is a Hilbert space with

$$(u, v)_{H^k(\Omega)} = \sum_{|\alpha| \le k} (D^\alpha u, D^\alpha v)_{L^2(\Omega)}.$$

In particular,

$$H^1(\Omega) = \left\{ v \in L^2(\Omega) : \nabla v \in L^2(\Omega)^d \right\}, \qquad \|v\|^2_{H^1(\Omega)} = \|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)} .$$

9.7 Traces, zero boundary values, and Poincaré's inequality

For a bounded Lipschitz domain, pointwise restriction to the boundary extends to a bounded linear trace operator

$$\mathrm{Tr} : H^1(\Omega) \longrightarrow H^{1/2}(\partial\Omega). \tag{9.3}$$

The space with homogeneous Dirichlet trace is

$$H^1_0(\Omega) = \overline{C^\infty_c(\Omega)}^{H^1(\Omega)} = \ker(\mathrm{Tr}).$$

Theorem 9.6 (Poincaré inequality)**.** *If Ω is bounded, connected, and Lipschitz, then there is a constant $C_P > 0$, depending only on Ω, such that*

$$\|v\|_{L^2(\Omega)} \le C_P \|\nabla v\|_{L^2(\Omega)} \qquad \forall v \in H^1_0(\Omega). \tag{9.4}$$

Consequently,

$$(u, v)_V = \int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}x, \qquad \|v\|_V = \|\nabla v\|_{L^2(\Omega)} ,$$

define an inner product and an equivalent norm on $V = H^1_0(\Omega)$.

Boundary values in Sobolev spaces

An H^1-function is an equivalence class defined almost everywhere in Ω, so the phrase $v = 0$ on $\partial\Omega$ is interpreted as $\operatorname{Tr} v = 0$. It is not a pointwise statement unless additional regularity is available.

Exercises

Exercise 9.1 (Finite-measure embedding)**.** Prove (9.1) for $q < \infty$ by applying Hölder's inequality to $|v|^p \cdot 1$. Treat $q = \infty$ separately, and construct an example showing that the inclusion can fail when $|\Omega| = \infty$.

Exercise 9.2 (A functional represented in L^2)**.** For fixed $g \in L^2(\Omega)$, set $F(v) = \int_\Omega gv \, \mathrm{d}x$. Show that $F \in (L^2(\Omega))'$ and that $\|F\|_{(L^2)'} = \|g\|_{L^2}$.

Exercise 9.3 (Weak and classical derivatives)**.** Let $u \in C^1(\Omega)$. Use integration by parts against test functions to show that its distributional derivative agrees with its classical derivative.

Exercise 9.4 (A jump is not in H^1)**.** Use the Heaviside example to show that a one-dimensional function with a nonzero jump cannot belong to H^1 on an interval containing the jump.

10 A general elliptic boundary-value problem

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. We consider the second-order operator

$$\mathcal{L}u = -\nabla \cdot (A\nabla u) + b \cdot \nabla u + cu, \tag{10.1}$$

where

$$A \in L^\infty(\Omega; \mathbb{R}^{d\times d}), \qquad b \in L^\infty(\Omega; \mathbb{R}^d), \qquad c \in L^\infty(\Omega).$$

Definition 10.1 (Uniform ellipticity)**.** The principal part is uniformly elliptic if there is $\alpha > 0$ such that

$$\xi^\top A(x)\xi \geq \alpha \, |\xi|^2 \qquad \forall \xi \in \mathbb{R}^d \tag{10.2}$$

for almost every $x \in \Omega$.

Only the symmetric part of A contributes to the quadratic form $\xi^\top A\xi$. Symmetry of A is not required for the weak problem, but it will matter for energy minimization.

In two dimensions, for a symmetric principal matrix

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

positive definiteness is equivalent to $a > 0$ and $ac - b^2 > 0$. In the classical notation $au_{xx} + 2bu_{xy} + cu_{yy}$, this is the condition $b^2 - ac < 0$. Uniform ellipticity requires the lower bound in (10.2), not merely pointwise positivity.

One convention for the convection term

The expressions $b \cdot \nabla u$ and $\nabla \cdot (bu)$ are not the same:

$$\nabla \cdot (bu) = b \cdot \nabla u + (\nabla \cdot b)u.$$

To agree with the preceding chapter, we use the advective form $b \cdot \nabla u$ throughout. Rewriting the operator in conservative form changes the effective zeroth-order coefficient.

10.1 Homogeneous Dirichlet conditions

Consider

$$\mathcal{L}u = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{10.3}$$

For smooth u and v with $v = 0$ on $\partial\Omega$, integration by parts gives

$$\int_\Omega (-\nabla \cdot (A\nabla u))v \,\mathrm{d}x = \int_\Omega A\nabla u \cdot \nabla v \,\mathrm{d}x.$$

This motivates the bilinear form

$$a(u, v) = \int_\Omega A\nabla u \cdot \nabla v \,\mathrm{d}x + \int_\Omega (b \cdot \nabla u)v \,\mathrm{d}x + \int_\Omega cuv \,\mathrm{d}x \tag{10.4}$$

on $V = H^1_0(\Omega)$. For $f \in L^2(\Omega)$, set

$$F(v) = \int_\Omega fv \,\mathrm{d}x.$$

More generally, the natural datum is any $F \in V' = H^{-1}(\Omega)$.

Definition 10.2 (Weak Dirichlet problem)**.** Find $u \in V = H^1_0(\Omega)$ such that

$$a(u, v) = \langle F, v \rangle_{V',V} \qquad \forall v \in V. \tag{10.5}$$

The form is continuous on $H^1(\Omega) \times H^1(\Omega)$. Indeed,

$$\begin{aligned} |a(w, v)| &\le \|A\|_{L^\infty} \|\nabla w\|_{L^2} \|\nabla v\|_{L^2} + \|b\|_{L^\infty} \|\nabla w\|_{L^2} \|v\|_{L^2} \\ &\quad + \|c\|_{L^\infty} \|w\|_{L^2} \|v\|_{L^2} \\ &\le M \|w\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)} \end{aligned} \tag{10.6}$$

for a coefficient-dependent constant M.

Exercises

Exercise 10.1 (Derivation of the weak form)**.** Starting from (10.1), derive (10.4) for smooth functions. Identify precisely the boundary term that disappears under a homogeneous Dirichlet condition.

Exercise 10.2 (Conservative versus advective convection)**.** Rewrite $-\nabla \cdot (A\nabla u) + \nabla \cdot (bu) + \widetilde{c}\,u$ in the form (10.1). Determine the relation between c and $\widetilde{c}$, and derive the corresponding weak form.

11 Distributional, weak, strong, and classical solutions

The word *solution* must specify both the function space and the meaning of the differential equation.

Definition 11.1 (Distributional solution)**.** A function $u \in H^1_{\mathrm{loc}}(\Omega)$ is a distributional solution of $\mathcal{L}u = f$ if

$$\langle \mathcal{L}u, \varphi \rangle = \langle f, \varphi \rangle \qquad \forall \varphi \in \mathcal{D}(\Omega),$$

where

$$\langle \mathcal{L}u, \varphi \rangle = \int_\Omega A\nabla u \cdot \nabla \varphi \, \mathrm{d}x + \int_\Omega (b \cdot \nabla u)\varphi \, \mathrm{d}x + \int_\Omega cu\varphi \, \mathrm{d}x.$$

Definition 11.2 (Strong and classical solutions)**.** A strong solution has enough Sobolev regularity for $\mathcal{L}u = f$ to hold almost everywhere, with all terms belonging to the prescribed data space. For sufficiently regular coefficients, a typical strong Dirichlet solution lies in $H^2(\Omega) \cap H^1_0(\Omega)$.

A classical solution is sufficiently smooth for the equation and boundary conditions to hold pointwise, for example $u \in C^2(\Omega) \cap C(\overline{\Omega})$ for a second-order Dirichlet problem with sufficiently smooth coefficients.

Proposition 11.3 (Weak and distributional formulations)**.** *Let $u \in H^1_0(\Omega)$, and assume that the two sides of (10.5) are continuous functionals on $H^1_0(\Omega)$. Then u solves (10.5) if and only if $\mathcal{L}u = F$ in $\mathcal{D}'(\Omega)$.*

Proof. If (10.5) holds, test it with $\varphi \in \mathcal{D}(\Omega) \subset H^1_0(\Omega)$. The definition of the distributional divergence gives

$$\langle \mathcal{L}u, \varphi \rangle = a(u, \varphi) = \langle F, \varphi \rangle .$$

Conversely, if this identity holds for every $\varphi \in \mathcal{D}(\Omega)$, use the density of $\mathcal{D}(\Omega)$ in $H^1_0(\Omega)$ and continuity to extend it to every $v \in H^1_0(\Omega)$. □

Hierarchy of solution concepts

Under the appropriate coefficient assumptions,

$$\boxed{\text{classical}} \Longrightarrow \boxed{\text{strong}} \Longrightarrow \boxed{\text{weak/distributional}}.$$

Reverse implications are regularity results and are not automatic.

For example, if $-\Delta u = f$, $f \in L^2(\Omega)$, and elliptic regularity gives $u \in H^2(\Omega)$, then the weak solution is also a strong solution. To conclude that it is classical one still needs an embedding into an appropriate space of continuously differentiable functions, which requires further regularity and depends on the dimension.

Terminology

Equality in $\mathcal{D}'(\Omega)$ defines a distributional solution, not by itself a strong solution. Calling it *strong* would erase the important distinction between weak existence and elliptic regularity.

Exercises

Exercise 11.1 (The Poisson equation in distributions)**.** Let $u \in H_0^1(\Omega)$. Show that

$$\int_\Omega \nabla u \cdot \nabla \varphi \, \mathrm{d}x = \int_\Omega f\varphi \, \mathrm{d}x \quad \forall \varphi \in \mathcal{D}(\Omega)$$

is equivalent to $-\Delta u = f$ in $\mathcal{D}'(\Omega)$.

Exercise 11.2 (Why density is needed)**.** Identify the density argument in the converse part of the proposition. Explain why equality only for smooth compactly supported test functions cannot be extended to $H_0^1(\Omega)$ without continuity.

12 Well-posedness: the Lax–Milgram theorem

Theorem 12.1 (Lax–Milgram)**.** *Let V be a real Hilbert space. Suppose that $a : V \times V \to \mathbb{R}$ is bilinear and that there are constants $M, \alpha_0 > 0$ such that*

$$|a(w,v)| \le M \|w\|_V \|v\|_V \qquad \forall w, v \in V, \tag{12.1}$$

$$a(v,v) \ge \alpha_0 \|v\|_V^2 \qquad \forall v \in V. \tag{12.2}$$

Then, for every $F \in V'$, there is a unique $u \in V$ such that

$$a(u,v) = F(v) \qquad \forall v \in V.$$

Moreover,

$$\|u\|_V \le \frac{1}{\alpha_0} \|F\|_{V'}. \tag{12.3}$$

The estimate follows immediately by testing with $v = u$:

$$\alpha_0 \|u\|_V^2 \le a(u,u) = F(u) \le \|F\|_{V'} \|u\|_V.$$

It expresses continuous dependence on the data and is therefore a stability estimate.

The symmetric case

If a is also symmetric, then $(w,v)_a = a(w,v)$ is an inner product whose norm is equivalent to $\|\cdot\|_V$. The Riesz representation theorem in this new Hilbert structure gives the solution directly. The nonsymmetric Lax–Milgram theorem is a genuine extension of this argument.

12.1 Coercivity of the elliptic form

Assume now that $b \in W^{1,\infty}(\Omega; \mathbb{R}^d)$. For $v \in H_0^1(\Omega)$, integration by parts gives

$$\int_\Omega (b \cdot \nabla v) v \, \mathrm{d}x = \frac{1}{2} \int_\Omega b \cdot \nabla (v^2) \, \mathrm{d}x = -\frac{1}{2} \int_\Omega (\nabla \cdot b) v^2 \, \mathrm{d}x. \tag{12.4}$$

There is no boundary contribution because $\operatorname{Tr} v = 0$. Combining (10.2) and (12.4) yields

$$a(v,v) \ge \alpha \|\nabla v\|_{L^2}^2 + \int_\Omega \left(c - \frac{1}{2} \nabla \cdot b \right) v^2 \, \mathrm{d}x. \tag{12.5}$$

Thus the sufficient condition

$$c - \frac{1}{2}\nabla \cdot b \geq 0 \qquad \text{almost everywhere in } \Omega \tag{12.6}$$

implies

$$a(v,v) \geq \alpha \left\|\nabla v\right\|_{L^2}^2 .$$

Poincaré's inequality then gives coercivity in the full H^1-norm:

$$a(v,v) \geq \frac{\alpha}{1+C_P^2} \left\|v\right\|_{H^1(\Omega)}^2 . \tag{12.7}$$

Theorem 12.2 (Well-posed homogeneous Dirichlet problem)**.** *Assume* (10.2) *and* (12.6)*. Then, for every $F \in H^{-1}(\Omega)$, the weak problem* (10.5) *has a unique solution $u \in H_0^1(\Omega)$, and*

$$\left\|u\right\|_{H^1(\Omega)} \leq \frac{1+C_P^2}{\alpha} \left\|F\right\|_{H^{-1}(\Omega)} .$$

If the energy norm $\left\|v\right\|_V = \left\|\nabla v\right\|_{L^2}$ is used instead, the corresponding coercivity constant is simply α.

Remark 12.3. Condition (12.6) is convenient and sufficient, but it is not necessary for well-posedness. More general problems can be treated using a Gårding inequality together with a uniqueness argument or Fredholm theory.

Exercises

Exercise 12.1 (Boundedness of the load)**.** For $f \in L^2(\Omega)$, show that $F(v) = \int_\Omega f v \,\mathrm{d}x$ belongs to $H^{-1}(\Omega)$ and obtain an explicit bound for $\left\|F\right\|_{H^{-1}}$ in terms of C_P and $\left\|f\right\|_{L^2}$.

Exercise 12.2 (Coercivity with a negative part)**.** Suppose $c - \frac{1}{2}\nabla \cdot b \geq -\eta$. Use Poincaré's inequality to find a sufficient upper bound on η that still guarantees coercivity.

13 Dirichlet, Neumann, Robin, and mixed conditions

For a smooth solution of $\mathcal{L}u = f$, integration by parts gives

$$a(u,v) = \int_\Omega f v \,\mathrm{d}x + \int_{\partial\Omega} (A\nabla u \cdot n) v \,\mathrm{d}s. \tag{13.1}$$

The test and trial spaces determine how the boundary term is treated.

13.1 Homogeneous and nonhomogeneous Dirichlet data

For $u = 0$ on all of $\partial\Omega$, take $V = H_0^1(\Omega)$; the trace of every test function vanishes, so the boundary term in (13.1) disappears.

If $u = g_D$ on $\partial\Omega$, the trial set is affine rather than linear:

$$V_{g_D} = \left\{ v \in H^1(\Omega) : \operatorname{Tr} v = g_D \right\}.$$

The trace theorem has a bounded right inverse

$$R : H^{1/2}(\partial\Omega) \longrightarrow H^1(\Omega), \qquad \operatorname{Tr}(Rg) = g.$$

Choose a lifting $u_g = Rg_D$ and write

$$u = u_0 + u_g, \qquad u_0 \in H^1_0(\Omega).$$

Then u_0 solves

$$a(u_0, v) = F(v) - a(u_g, v) \qquad \forall v \in H^1_0(\Omega). \tag{13.2}$$

13.2 Neumann conditions

For the conormal Neumann condition

$$A\nabla u \cdot n = g_N \quad \text{on } \partial\Omega,$$

the natural space is $V = H^1(\Omega)$, and the weak problem is

$$a(u, v) = \int_\Omega fv \,\mathrm{d}x + \int_{\partial\Omega} g_N v \,\mathrm{d}s \qquad \forall v \in H^1(\Omega). \tag{13.3}$$

The trace theorem makes the boundary functional continuous under suitable regularity of g_N, for example $g_N \in L^2(\partial\Omega)$.

For the pure Poisson problem, constants lie in the kernel. Testing with $v = 1$ gives the compatibility condition

$$\int_\Omega f \,\mathrm{d}x + \int_{\partial\Omega} g_N \,\mathrm{d}s = 0. \tag{13.4}$$

When it holds, the solution is determined only up to an additive constant; uniqueness can be restored by imposing $\int_\Omega u \,\mathrm{d}x = 0$.

13.3 Robin conditions

For

$$A\nabla u \cdot n + \mu u = g_R \quad \text{on } \partial\Omega,$$

the weak problem on $H^1(\Omega)$ becomes

$$a(u, v) + \int_{\partial\Omega} \mu uv \,\mathrm{d}s = \int_\Omega fv \,\mathrm{d}x + \int_{\partial\Omega} g_R v \,\mathrm{d}s. \tag{13.5}$$

If $\mu \geq 0$ and is positive on a sufficiently large part of the boundary, the boundary term can remove the constant kernel and contribute to coercivity.

13.4 Mixed conditions and lifting

Suppose

$$\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}, \qquad \Gamma_D \cap \Gamma_N = \varnothing,$$

and consider

$$\mathcal{L}u = f \text{ in } \Omega, \qquad u = g_D \text{ on } \Gamma_D, \qquad A\nabla u \cdot n = g_N \text{ on } \Gamma_N.$$

The test space and affine trial space are

$$V_0 = \left\{ v \in H^1(\Omega) : \operatorname{Tr} v = 0 \text{ on } \Gamma_D \right\},$$
$$V_{g_D} = \left\{ v \in H^1(\Omega) : \operatorname{Tr} v = g_D \text{ on } \Gamma_D \right\}.$$

The weak problem is

$$\text{find } u \in V_{g_D} \text{ such that} \qquad a(u,v) = \int_\Omega fv \,\mathrm{d}x + \int_{\Gamma_N} g_N v \,\mathrm{d}s \quad \forall v \in V_0. \tag{13.6}$$

If g_D is the restriction of an admissible $H^{1/2}$-trace, choose a lifting $u_g \in H^1(\Omega)$ with $\operatorname{Tr} u_g = g_D$ on Γ_D. Writing $u = u_0 + u_g$ reduces (13.6) to a problem for $u_0 \in V_0$:

$$a(u_0, v) = \int_\Omega fv \,\mathrm{d}x + \int_{\Gamma_N} g_N v \,\mathrm{d}s - a(u_g, v) \qquad \forall v \in V_0. \tag{13.7}$$

If Γ_D has positive boundary measure, a Poincaré inequality holds on V_0, providing the same coercivity mechanism as in the full Dirichlet problem.

Essential and natural boundary conditions

Dirichlet data are *essential*: they are imposed through the trial and test spaces. Neumann data are *natural*: they enter the load functional after integration by parts. Robin data modify both the bilinear form and the load.

Exercises

Exercise 13.1 (Pure Neumann compatibility)**.** Derive (13.4) both from the weak formulation and by integrating $-\Delta u = f$ over Ω. Explain why the two derivations give the same sign.

Exercise 13.2 (Independence of the lifting)**.** Let u_g and $\widetilde{u}_g$ be two liftings of the same Dirichlet datum. Show that the corresponding homogeneous unknowns differ, but the reconstructed solution u is the same.

Exercise 13.3 (A Robin coercivity estimate)**.** For the Poisson equation with $\mu \geq \mu_0 > 0$ on $\partial\Omega$, show that

$$\int_\Omega |\nabla v|^2 \,\mathrm{d}x + \int_{\partial\Omega} \mu v^2 \,\mathrm{d}s$$

defines a norm equivalent to the $H^1(\Omega)$-norm. You may use an appropriate trace–Poincaré inequality.

14 Elliptic regularity

Lax–Milgram gives a solution in the energy space, typically H^1. Additional smoothness of the solution is a separate theorem and depends on the coefficients, the data, the domain, and the boundary conditions.

Theorem 14.1 (Basic H^2-regularity for Poisson)**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded $C^{1,1}$ domain, or a convex polygon when $d = 2$. If $f \in L^2(\Omega)$ and $u \in H^1_0(\Omega)$ solves*

$$-\Delta u = f,$$

then $u \in H^2(\Omega)$ and

$$\|u\|_{H^2(\Omega)} \leq C_{\mathrm{reg}} \|f\|_{L^2(\Omega)}. \tag{14.1}$$

On sufficiently smooth domains, with smooth compatible coefficients and boundary data, this principle extends to estimates of the form

$$\|u\|_{H^{s+2}(\Omega)} \leq C \|f\|_{H^s(\Omega)}.$$

The precise hypotheses become more restrictive as s increases.

14.1 Corner singularities

Regularity can fail on a nonsmooth domain even when f is smooth. Near a corner of opening angle ω, the model Dirichlet singular function is

$$u_s(r,\theta) = r^{\lambda}\sin(\lambda\theta), \qquad \lambda = \frac{\pi}{\omega}.$$

If $\omega > \pi$, then $\lambda < 1$, and in general

$$u_s \in H^{1+s} \quad \text{for every } s < \lambda, \qquad u_s \notin H^2.$$

Thus nonconvex corners can prevent the two-derivative gain in (14.1). Problems with different boundary conditions on adjacent boundary portions can also develop singularities at the transition points, even when the geometric boundary is otherwise smooth.

Why regularity matters numerically

Galerkin quasi-optimality compares the numerical error with the best approximation error. The latter is estimated by interpolation theory and therefore depends on Sobolev regularity. A method may be stable and quasi-optimal while converging more slowly because the exact solution has a corner or mixed-boundary singularity.

Exercises

Exercise 14.1 (Integrability of a corner singularity)**.** For $u_s(r,\theta) = r^{\lambda}\sin(\lambda\theta)$, estimate derivatives of order m near $r = 0$. Use polar coordinates to determine for which integer m these derivatives belong to L^2.

Exercise 14.2 (Weak versus strong on a convex polygon)**.** Let $\Omega \subset \mathbb{R}^2$ be a convex polygon and $f \in L^2(\Omega)$. Combine Lax–Milgram and (14.1) to explain why the weak Dirichlet solution of the Poisson problem is also a strong solution.

15 The Galerkin method

Let V be a Hilbert space, let $a : V \times V \to \mathbb{R}$ satisfy the continuity and coercivity assumptions (12.1)–(12.2), and let $F \in V'$. The continuous problem is

$$\text{find } u \in V \text{ such that} \qquad a(u,v) = F(v) \quad \forall v \in V. \tag{15.1}$$

Choose a finite-dimensional conforming subspace $V_h \subset V$. The Galerkin problem is

$$\text{find } u_h \in V_h \text{ such that} \qquad a(u_h,v_h) = F(v_h) \quad \forall v_h \in V_h. \tag{15.2}$$

The finite element method is obtained by choosing V_h to consist of piecewise polynomial functions on a mesh. For now, no particular construction of V_h is needed.

15.1 Discrete well-posedness and stability

The same constants M and α_0 apply on the subspace V_h. Lax–Milgram therefore gives a unique Galerkin solution and the uniform estimate

$$\|u_h\|_V \le \frac{1}{\alpha_0}\,\|F\|_{V'}\,. \tag{15.3}$$

The constant is independent of $\dim V_h$; this is the stability of the conforming Galerkin method.

15.2 The algebraic system

Let $\{\phi_1, \dots, \phi_n\}$ be a basis of V_h, and write

$$u_h = \sum_{j=1}^{n} U_j \phi_j.$$

Testing with ϕ_i gives

$$\sum_{j=1}^{n} a(\phi_j, \phi_i) U_j = F(\phi_i), \qquad i = 1, \dots, n.$$

Thus

$$\mathbf{AU} = \mathbf{F}, \qquad A_{ij} = a(\phi_j, \phi_i), \qquad F_i = F(\phi_i). \tag{15.4}$$

Coercivity makes $\mathbf{A}$ nonsingular. If a is symmetric, then $\mathbf{A}$ is symmetric positive definite.

15.3 Energy minimization in the symmetric case

Suppose a is symmetric. Define

$$J(v) = \frac{1}{2} a(v, v) - F(v).$$

The continuous solution is the unique minimizer of J over V, while the Galerkin solution is the unique minimizer over V_h:

$$J(u) = \min_{v \in V} J(v), \qquad J(u_h) = \min_{v_h \in V_h} J(v_h).$$

At the matrix level, $\mathbf{U}$ minimizes

$$J_h(\mathbf{V}) = \frac{1}{2} \mathbf{V}^\top \mathbf{A} \mathbf{V} - \mathbf{V}^\top \mathbf{F}.$$

15.4 Galerkin orthogonality

Proposition 15.1 (Galerkin orthogonality)**.** *The error $e_h = u - u_h$ satisfies*

$$a(e_h, v_h) = 0 \qquad \forall v_h \in V_h. \tag{15.5}$$

Proof. Test the continuous problem with $v_h \in V_h$ and subtract the discrete equation:

$$a(u - u_h, v_h) = F(v_h) - F(v_h) = 0.$$

□

This is the exact consistency property of a conforming Galerkin method: the continuous solution satisfies every discrete test equation.

15.5 Céa's lemma and convergence

Theorem 15.2 (Céa's lemma)**.** *Under the Lax–Milgram assumptions,*

$$\|u - u_h\|_V \leq \frac{M}{\alpha_0} \inf_{v_h \in V_h} \|u - v_h\|_V . \tag{15.6}$$

Proof. Let $v_h \in V_h$ be arbitrary. By coercivity and Galerkin orthogonality,

$$\begin{aligned} \alpha_0 \|u - u_h\|_V^2 &\leq a(u - u_h, u - u_h) \\ &= a(u - u_h, u - v_h) + a(u - u_h, v_h - u_h) \\ &= a(u - u_h, u - v_h) \\ &\leq M \|u - u_h\|_V \|u - v_h\|_V . \end{aligned}$$

Cancel $\|u - u_h\|_V$ and take the infimum over $v_h \in V_h$. □

Céa's lemma says that a conforming Galerkin solution is quasi-optimal: its error is bounded by a fixed multiple of the best error available in the chosen space. Therefore,

$$\inf_{v_h \in V_h} \|u - v_h\|_V \longrightarrow 0 \quad \Longrightarrow \quad \|u - u_h\|_V \longrightarrow 0. \tag{15.7}$$

Proposition 15.3 (Exact best approximation in the energy norm)**.** *If a is symmetric, define $\|v\|_a = \sqrt{a(v,v)}$. Then u_h is the a-orthogonal projection of u onto V_h, and*

$$\|u - u_h\|_a = \min_{v_h \in V_h} \|u - v_h\|_a . \tag{15.8}$$

Consequently,

$$\|u - u_h\|_V \leq \sqrt{\frac{M}{\alpha_0}} \inf_{v_h \in V_h} \|u - v_h\|_V . \tag{15.9}$$

Proof. Galerkin orthogonality is now orthogonality in the inner product a. For any $v_h \in V_h$, the Pythagorean identity gives

$$\|u - v_h\|_a^2 = \|u - u_h\|_a^2 + \|u_h - v_h\|_a^2 .$$

This proves (15.8). The norm equivalence

$$\sqrt{\alpha_0} \|v\|_V \leq \|v\|_a \leq \sqrt{M} \|v\|_V$$

then yields (15.9). □

The Galerkin proof pattern

$$\boxed{\text{Lax–Milgram stability}} + \boxed{\text{Galerkin orthogonality}} + \boxed{\text{approximation by } V_h} \Longrightarrow \boxed{\text{convergence}}.$$

The forthcoming finite element analysis will concentrate on the third box: constructing local polynomial spaces and estimating their best approximation error.

15.6 A theoretical dense sequence

If V is separable, choose a complete orthonormal system $\{\phi_j\}_{j\geq 1}$ and set

$$V_n = \operatorname{span}\{\phi_1, \ldots, \phi_n\}.$$

Then $\bigcup_n V_n$ is dense in V, and Céa's lemma proves convergence of the corresponding Galerkin solutions. This construction establishes the abstract principle but is rarely convenient computationally. Finite elements replace global basis functions with locally supported piecewise polynomials; spectral methods instead use global high-order functions.

Exercises

Exercise 15.1 (Matrix coercivity)**.** Starting from (15.4), show directly that $\mathbf{AU} = 0$ implies $\mathbf{U} = 0$. If a is symmetric, prove that $\mathbf{A}$ is symmetric positive definite.

Exercise 15.2 (Galerkin minimization)**.** Assume that a is symmetric. Differentiate $J_h(\mathbf{V}) = \frac{1}{2}\mathbf{V}^\top \mathbf{AV} - \mathbf{V}^\top \mathbf{F}$ and show that its unique stationary point is the Galerkin coefficient vector. Explain why this point is the global minimizer.

Exercise 15.3 (Nested spaces)**.** Suppose $V_1 \subset V_2 \subset \cdots \subset V$ and the union is dense in V. Use Céa's lemma to prove convergence. In the symmetric case, show that the energy errors $\|u - u_n\|_a$ form a nonincreasing sequence.

Exercise 15.4 (A first finite element preview)**.** On $\Omega = (0, 1)$, let V_h be the space of continuous piecewise affine functions on a partition, vanishing at both endpoints. For $a(u, v) = \int_0^1 u'v'\,\mathrm{d}x$, compute the entries associated with two neighboring hat functions and identify the sparsity pattern of the matrix.

16 Finite element spaces in one dimension

The abstract Galerkin theory of section 15 reduces the analysis of a conforming method to an approximation question: how well can a finite-dimensional subspace approximate the exact solution? Finite elements answer this question with piecewise polynomials having small support. We first make the construction precise in one dimension, where all of the main ideas are already visible.

16.1 A model reaction–diffusion problem

Let $\Omega = (a,b)$, and consider

$$-(\kappa u')' + cu = f \quad \text{in } (a,b), \qquad u(a) = u(b) = 0, \tag{16.1}$$

where $\kappa, c \in L^\infty(a,b)$,

$$\kappa(x) \geq \kappa_0 > 0, \qquad c(x) \geq 0 \quad \text{for almost every } x \in (a,b), \tag{16.2}$$

and $f \in L^2(a,b)$. With $V = H^1_0(a,b)$, set

$$a(w,v) = \int_a^b (\kappa w' v' + cwv)\,\mathrm{d}x, \qquad F(v) = \int_a^b fv\,\mathrm{d}x. \tag{16.3}$$

The weak problem is to find $u \in V$ such that $a(u,v) = F(v)$ for every $v \in V$. Under (16.2), Poincaré's inequality and Lax–Milgram give a unique solution.

Choose a partition

$$a = x_0 < x_1 < \cdots < x_N = b, \qquad K_i = (x_{i-1}, x_i), \qquad h_i = x_i - x_{i-1}, \qquad h = \max_i h_i. \tag{16.4}$$

For an integer $k \geq 1$, define the conforming finite element space

$$V_h^k = \left\{ v_h \in C^0([a,b]) : v_h|_{K_i} \in \mathbb{P}_k(K_i),\ i = 1,\dots,N,\ v_h(a) = v_h(b) = 0 \right\}. \tag{16.5}$$

Here $\mathbb{P}_k(K)$ denotes the polynomials of degree at most k on K. Since a continuous piecewise polynomial has a square-integrable weak derivative, $V_h^k \subset H^1_0(a,b)$.

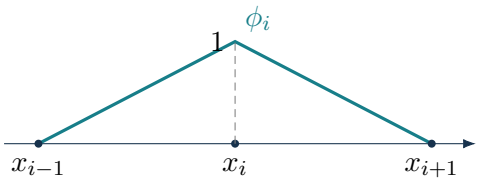
The finite element problem is

$$\text{find } u_h \in V_h^k \text{ such that} \qquad a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h^k. \tag{16.6}$$

This is an instance of the conforming Galerkin method, so existence, uniqueness, Galerkin orthogonality, and Céa's lemma follow from the preceding chapter.

16.2 Degrees of freedom and local basis functions

For $k = 1$, the values at the mesh vertices determine a function in V_h^1. The interior nodal basis $\{\phi_i\}_{i=1}^{N-1}$ is characterized by $\phi_i(x_j) = \delta_{ij}$. Each ϕ_i is supported on $[x_{i-1}, x_{i+1}]$, which is the source of the sparse algebraic system.



For $k > 1$, choose $k+1$ distinct reference nodes $0 = \widehat{\xi}_0 < \cdots < \widehat{\xi}_k = 1$. Their images on every element are

$$x_{i,r} = x_{i-1} + h_i \widehat{\xi}_r, \qquad r = 0, \ldots, k.$$

Endpoint nodes are shared by neighboring elements, while the $k-1$ interior nodes of an element are local to it. Consequently,

$$\dim V_h^k = Nk - 1. \tag{16.7}$$

The global Lagrange basis is defined by requiring each basis function to take the value one at one global node and zero at all the others.

FEM stencil

A vertex basis function meets two elements; an element-interior basis function is supported on only one. Thus increasing k increases the local stencil but does not destroy sparsity.

Writing $u_h = \sum_{j=1}^{n} U_j \phi_j$, $n = Nk - 1$, gives

$$(\mathbf{K} + \mathbf{M})\mathbf{U} = \mathbf{F}, \qquad K_{ij} = \int_a^b \kappa \phi_j' \phi_i' \, \mathrm{d}x, \quad M_{ij} = \int_a^b c \phi_j \phi_i \, \mathrm{d}x, \quad F_i = \int_a^b f \phi_i \, \mathrm{d}x. \tag{16.8}$$

Only basis functions with overlapping supports interact, so both $\mathbf{K}$ and $\mathbf{M}$ are *banded*. They are called the *stiffness* and *mass* matrices, respectively.

16.3 Nodal interpolation and approximation

For a continuous function v, its degree-k nodal interpolant $I_h v \in V_h^k$ is specified by

$$(I_h v)(x_{i,r}) = v(x_{i,r}) \quad \text{at every global Lagrange node.} \tag{16.9}$$

If $v(a) = v(b) = 0$, then $I_h v \in V_h^k$. The construction is local: $I_h v|_{K_i}$ depends only on $v|_{K_i}$.

Theorem 16.1 (One-dimensional interpolation estimate)**.** *Let $0 \le m \le s \le k+1$, with m, s integers and $s \ge 1$. There is a constant C, depending on k and the reference nodes but not on h_i, such that*

$$\|v - I_h v\|_{H^m(K_i)} \le C h_i^{s-m} \, |v|_{H^s(K_i)} \qquad \forall v \in H^s(K_i). \tag{16.10}$$

Whenever the nodal values in (16.9) are meaningful, summing over the mesh yields, for $m = 0, 1$,

$$\|v - I_h v\|_{H^m(a,b)} \le C h^{s-m} \, \|v\|_{H^s(a,b)} \, . \tag{16.11}$$

The local estimate follows by transporting the interpolation operator to the unit interval, proving boundedness there, and scaling back. The power h_i^{s-m} records the number of derivatives gained by approximation. For the maximal regularity $s = k+1$,

$$\|v - I_h v\|_{L^2(a,b)} \le C h^{k+1} \, \|v\|_{H^{k+1}(a,b)} \, , \qquad \|v - I_h v\|_{H^1(a,b)} \le C h^{k} \, \|v\|_{H^{k+1}(a,b)} \, . \tag{16.12}$$

If only $v \in H^s(a,b)$ with $1 < s < k+1$, the degree cannot create missing smoothness; the orders reduce to h^s in L^2 and h^{s-1} in H^1.

Two routes to higher accuracy

The error is governed jointly by the mesh size, the polynomial degree, and the regularity of the solution. One may decrease h (h-refinement), increase k (p-refinement), or combine both (hp-refinement). Raising k is effective only where the solution is smooth.

16.4 Energy- and L^2-error estimates

Applying Céa's lemma with $v_h = I_h u$ immediately gives the energy-norm estimate.

Theorem 16.2 (Finite element energy error in one dimension)**.** *Suppose $u \in H^s(a,b) \cap H_0^1(a,b)$, where $1 < s \le k+1$. Then the solution of (16.6) satisfies*

$$\|u - u_h\|_{H^1(a,b)} \le C h^{s-1} \|u\|_{H^s(a,b)} . \tag{16.13}$$

In particular, if $u \in H^{k+1}(a,b)$, the method has order k in the energy norm.

To recover one further power of h in L^2, let $e = u - u_h$, and solve the adjoint problem

$$a(v,z) = (e,v)_{L^2(a,b)} \qquad \forall v \in H_0^1(a,b). \tag{16.14}$$

For a symmetric bilinear form, this is the same differential operator with right-hand side e. If the dual solution satisfies

$$\|z\|_{H^2(a,b)} \le C_{\mathrm{reg}} \|e\|_{L^2(a,b)} , \tag{16.15}$$

then Galerkin orthogonality gives

$$\begin{aligned} \|e\|_{L^2}^2 &= a(e,z) = a(e, z - I_h z) \\ &\le C \|e\|_{H^1} \|z - I_h z\|_{H^1} \le Ch \|e\|_{H^1} \|z\|_{H^2} . \end{aligned} \tag{16.16}$$

After using (16.15) and cancelling one factor of $\|e\|_{L^2}$, we obtain

$$\|u - u_h\|_{L^2(a,b)} \le Ch \|u - u_h\|_{H^1(a,b)} \le Ch^s \|u\|_{H^s(a,b)} . \tag{16.17}$$

This is the *Aubin–Nitsche duality argument*. For a nonsymmetric operator, one uses the adjoint bilinear form in (16.14); the additional order still depends on regularity of the adjoint problem.

Regularity is part of the estimate

The formal rates h^k in H^1 and h^{k+1} in L^2 require $u \in H^{k+1}$, and the latter also uses dual regularity. Corners, discontinuous coefficients, or incompatible data may lower the Sobolev regularity and therefore the observed convergence rate.

Exercises

Exercise 16.1 (Quadratic degrees of freedom)**.** On a partition with N intervals, place one degree of freedom at each vertex and one at each midpoint. Count the unknowns after imposing homogeneous Dirichlet conditions, and describe the support of the two types of basis function (vertex-based and midpoint-based).

Exercise 16.2 (Local matrices)**.** On an interval $K = (x_{i-1}, x_i)$, compute the degree-one element stiffness and mass matrices for constant κ and c. Assemble their contribution into the two corresponding global rows.

17 Simplicial finite elements

We now pass to polygonal or polyhedral domains in dimension $d \in \{2, 3\}$. The main new ingredients are the geometry of the mesh, the compatibility of traces across element interfaces, and the systematic assembly of element contributions.

17.1 Triangulations and polynomial spaces

Let $\Omega \subset \mathbb{R}^d$ be a bounded polygonal or polyhedral domain. A *simplicial triangulation* $\mathcal{T}_h$ of $\overline{\Omega}$ is a finite collection of closed triangles when $d = 2$, or tetrahedra when $d = 3$, such that

1. $\overline{\Omega} = \bigcup_{T \in \mathcal{T}_h} T$;
2. the interiors of distinct elements are disjoint;
3. the intersection of two distinct elements is either empty or a complete common subsimplex: a vertex, an edge, and in three dimensions possibly a face.

The last requirement excludes *hanging nodes* (vertices of one element belonging to the interior of an edge/face of a neighbouring element) for the moment and makes the mesh *conforming* (*regular*). We write

$$h_T = \operatorname{diam} T, \qquad h = \max_{T \in \mathcal{T}_h} h_T, \qquad \rho_T = \text{diameter of the largest ball contained in } T. \tag{17.1}$$

Definition 17.1 (Shape regularity and quasi-uniformity)**.** A family $\{\mathcal{T}_h\}_h$ is *shape regular* if there is a constant σ, independent of h, such that

$$\frac{h_T}{\rho_T} \le \sigma \qquad \forall T \in \mathcal{T}_h. \tag{17.2}$$

It is *quasi-uniform* if, in addition, there is C_{qu} such that $h \le C_{\mathrm{qu}} h_T$ for every $T \in \mathcal{T}_h$.

Shape regularity prevents elements from becoming arbitrarily flat; it does not require all elements to have comparable size. Quasi-uniformity is the stronger, global size condition used later in Section 19 for the simplest estimate of the conditioning of the linear systems resulting from the FEM.

For a simplex T, let

$$\mathbb{P}_k(T) = \{p : T \to \mathbb{R} : p \text{ is a polynomial of total degree at most } k\}. \tag{17.3}$$

Its dimension is

$$\dim \mathbb{P}_k(T) = \binom{k+d}{d}. \tag{17.4}$$

The conforming Lagrange space is

$$V_h^k = \left\{ v_h \in C^0(\overline{\Omega}) : v_h|_T \in \mathbb{P}_k(T) \quad \forall T \in \mathcal{T}_h \right\}, \qquad V_{h,0}^k = V_h^k \cap H_0^1(\Omega). \tag{17.5}$$

Continuity has a precise Sobolev interpretation.

Proposition 17.2 (Piecewise H^1 characterization)**.** *Suppose $v \in L^2(\Omega)$ and $v|_T \in H^1(T)$ for every $T \in \mathcal{T}_h$. Then $v \in H^1(\Omega)$ if and only if the two traces of v agree almost everywhere on every interior face.*

For piecewise polynomials, equality of the traces is classical equality on the face. Thus the global continuity in (17.5) is exactly what removes distributional jump terms from the first derivatives. Consequently $V_h^k \subset H^1(\Omega)$.

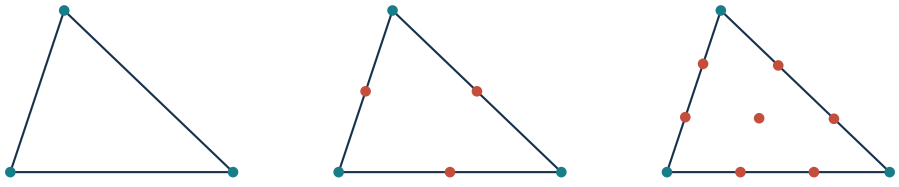


Figure 1: The degree-k Lagrange nodes corresponding to $k = 1, 2, 3$.

17.2 Lagrange nodes on a simplex

Let the vertices of a d-simplex T be $a_0, \dots, a_d$. The degree-k Lagrange nodes are

$$\mathcal{N}_k(T) = \left\{ x \in T : x = \frac{1}{k} \sum_{i=0}^{d} \alpha_i a_i, \quad \alpha_i \in \mathbb{N}_0, \quad \sum_{i=0}^{d} \alpha_i = k \right\}. \tag{17.6}$$

For $k = 1$ these are the vertices; for $k = 2$ they are the vertices and edge midpoints; for $k \geq 3$ also nodes inside the element are required, as shown in Figure 17.2. The node values form a unisolvent set of degrees of freedom for $\mathbb{P}_k(T)$: a polynomial in $\mathbb{P}_k(T)$ is uniquely determined by its values on $\mathcal{N}_k(T)$. We refer to the fundamental book by Ciarlet for a proof and for a complete treatment of these elements and many more [4].

After geometrically identical nodes on adjacent elements have been identified, enumerate the global nodes by $z_1, \dots, z_n$. The global nodal basis $\{\phi_i\}_{i=1}^{n}$ satisfies

$$\phi_i(z_j) = \delta_{ij}. \tag{17.7}$$

Its support is the patch of elements containing z_i. Therefore $a(\phi_j, \phi_i) = 0$ whenever the supports of ϕ_i and ϕ_j do not overlap. Local support, rather than merely low polynomial degree, is the structural reason for matrix sparsity.

17.3 The reference simplex and affine maps

All element computations can be reduced to one reference simplex $\widehat{T}$. In two dimensions take

$$\widehat{T} = \{(\widehat{x}_1, \widehat{x}_2) : \widehat{x}_1 \geq 0,\ \widehat{x}_2 \geq 0,\ \widehat{x}_1 + \widehat{x}_2 \leq 1\}. \tag{17.8}$$

For each $T \in \mathcal{T}_h$, let

$$F_T(\widehat{x}) = B_T \widehat{x} + b_T \tag{17.9}$$

be the affine bijection from $\widehat{T}$ to T. If $v = \widehat{v} \circ F_T^{-1}$, then

$$\nabla v(x) = B_T^{-\top} \widehat{\nabla} \widehat{v}(\widehat{x}), \qquad \mathrm{d}x = |\det B_T| \, \mathrm{d}\widehat{x}. \tag{17.10}$$

Hence, for example,

$$\int_T \kappa(x) \nabla \phi_j \cdot \nabla \phi_i \, \mathrm{d}x = |\det B_T| \int_{\widehat{T}} \kappa(F_T(\widehat{x})) (B_T^{-\top} \widehat{\nabla} \widehat{\phi}_j) \cdot (B_T^{-\top} \widehat{\nabla} \widehat{\phi}_i) \, \mathrm{d}\widehat{x}. \tag{17.11}$$

The basis and quadrature data on $\widehat{T}$ can therefore be reused on every physical element; only B_T, b_T, and the coefficients of the PDE change.

17.4 Element matrices and global assembly

For the reaction–diffusion form

$$a(w,v)=\int_\Omega (\kappa\nabla w\cdot\nabla v + cwv)\,\mathrm{d}x,$$

define on each element the local matrix and vector

$$A^T_{rs}=\int_T (\kappa\nabla\phi^T_s\cdot\nabla\phi^T_r + c\phi^T_s\phi^T_r)\,\mathrm{d}x, \qquad F^T_r=\int_T f\phi^T_r\,\mathrm{d}x. \tag{17.12}$$

Let $\mathsf{g}_T(r)$ map the local degree of freedom r on T to its global index. Assembly consists of the additions

$$A_{\mathsf{g}_T(r),\mathsf{g}_T(s)} \mathrel{+}= A^T_{rs}, \qquad F_{\mathsf{g}_T(r)} \mathrel{+}= F^T_r, \tag{17.13}$$

for every element and every pair of local indices. The symbol $+=$ is essential: neighboring elements contribute to the same global entry.

In practice the integrals are evaluated by a quadrature rule on $\widehat{T}$,

$$\int_{\widehat{T}} g(\widehat{x})\,\mathrm{d}\widehat{x} \approx \sum_{q=1}^{N_q} \widehat{w}_q g(\widehat{x}_q). \tag{17.14}$$

For constant coefficients and degree-k Lagrange elements, the stiffness integrand is a polynomial of degree at most $2k-2$, while the mass integrand has degree at most $2k$. Exact integration of both terms therefore requires a rule exact through degree $2k$. Variable coefficients or nonpolynomial data generally require approximate quadrature.

Quadrature changes the variational problem

With inexact quadrature, the computed bilinear form need not equal $a(\cdot,\cdot)$ on V_h^k. The resulting method is a *variational crime*; it can still converge optimally when the quadrature error is controlled, but exact Galerkin orthogonality must be replaced by a consistency estimate.

17.5 Essential boundary conditions

For homogeneous Dirichlet data, one may construct a basis only for interior degrees of freedom. For nonhomogeneous nodal data, split the coefficient vector into free and prescribed parts,

$$\mathbf{U}=\begin{pmatrix}\mathbf{U}_I\\ \mathbf{U}_D\end{pmatrix}, \qquad \begin{pmatrix}\mathbf{A}_{II} & \mathbf{A}_{ID}\\ \mathbf{A}_{DI} & \mathbf{A}_{DD}\end{pmatrix}\begin{pmatrix}\mathbf{U}_I\\ \mathbf{U}_D\end{pmatrix}=\begin{pmatrix}\mathbf{F}_I\\ \mathbf{F}_D\end{pmatrix}.$$

The unknown interior coefficients satisfy

$$\mathbf{A}_{II}\mathbf{U}_I=\mathbf{F}_I-\mathbf{A}_{ID}\mathbf{U}_D. \tag{17.15}$$

This elimination preserves symmetry and positive definiteness. An equivalent full-size implementation sets prescribed rows and columns to zero, places ones on the corresponding diagonal, and first transfers the column contributions to the right-hand side. Replacing only the rows is easy to code but destroys symmetry.

Neumann data are natural: a boundary integral appears in the load after integration by parts. Robin data contribute both a boundary mass term to the bilinear form and a boundary load term.

Exercises

Exercise 17.1 (Dimension of $\mathbb{P}_k$)**.** Count the monomials of total degree at most k in dimensions two and three, and verify (17.4).

Exercise 17.2 (The linear triangle)**.** On $\widehat{T}$, compute the three degree-one Lagrange basis functions, their gradients, and the 3×3 element stiffness matrix for the Poisson equation.

Exercise 17.3 (Trace matching)**.** Let two triangles share an edge. Show that two degree-k polynomials agree along the whole edge if they agree at the $k+1$ Lagrange nodes on that edge. Deduce continuity of the assembled Lagrange basis.

Exercise 17.4 (Assembly by connectivity)**.** For two triangles forming a square, number the four vertices, write the two local-to-global maps, and assemble the sparsity pattern of the degree-one stiffness matrix. No integration is required.

18 Polynomial approximation on simplicial meshes

The one-dimensional interpolation estimates extend to simplices once the dependence on element geometry is made explicit. The proof has three parts: a reference-element estimate, scaling under an affine map, and a mesh-regularity assumption that keeps the constants uniform.

18.1 Scaling under affine maps

Let $F_T : \widehat{T} \to T$ be the affine map (17.9), and define $\widehat{v} = v \circ F_T$. For an integer $m \geq 0$, repeated use of the chain rule gives

$$|\widehat{v}|_{H^m(\widehat{T})} \leq C \, |\det B_T|^{-1/2} \, \|B_T\|^m \, |v|_{H^m(T)} , \tag{18.1}$$

and, in the reverse direction,

$$|v|_{H^m(T)} \leq C \, |\det B_T|^{1/2} \, \left\|B_T^{-1}\right\|^m \, |\widehat{v}|_{H^m(\widehat{T})} . \tag{18.2}$$

The constants depend only on m and d. The same relations, summed from derivative order zero through m, hold for the full Sobolev norms.

Element geometry enters through

$$\|B_T\| \leq C h_T, \qquad \left\|B_T^{-1}\right\| \leq C \rho_T^{-1}, \qquad c\rho_T^d \leq |\det B_T| \leq C h_T^d. \tag{18.3}$$

On a shape-regular family, h_T/ρ_T is uniformly bounded. Products of the two estimates above can then be expressed with a constant independent of the particular element.

Lemma 18.1 (Affine scaling)**.** *Let $0 \leq m \leq s$ be integers and let the mesh family be shape regular. If an operator on $\widehat{T}$ satisfies*

$$\left\|\widehat{v} - \widehat{\Pi}\widehat{v}\right\|_{H^m(\widehat{T})} \leq C \, |\widehat{v}|_{H^s(\widehat{T})} ,$$

then its affine transport Π_T satisfies

$$\|v - \Pi_T v\|_{H^m(T)} \leq C h_T^{s-m} \, |v|_{H^s(T)} . \tag{18.4}$$

The constant is uniform over the mesh family.

The statement is often remembered as a dimensional rule: every derivative costs one factor of h_T^{-1}, while the volume measure contributes $h_T^{d/2}$ to an L^2-based norm.

18.2 The Bramble–Hilbert mechanism

The key fact is that an approximation operator reproducing all polynomials of degree k has an error controlled only by derivatives of order $k+1$. A convenient abstract form is the following.

Theorem 18.2 (Bramble–Hilbert lemma)**.** *Let $D \subset \mathbb{R}^d$ be a bounded domain that is star-shaped with respect to a ball, and let $s \geq 1$ be an integer. If a bounded linear functional $L : H^s(D) \to \mathbb{R}$ vanishes on $\mathbb{P}_{s-1}(D)$, then*

$$|L(v)| \leq C \, |v|_{H^s(D)} \qquad \forall v \in H^s(D). \tag{18.5}$$

More generally, if $\Pi : H^s(D) \to \mathbb{P}_{s-1}(D)$ is bounded and reproduces $\mathbb{P}_{s-1}(D)$, then, for $0 \leq m \leq s$,

$$\|v - \Pi v\|_{H^m(D)} \leq C \, |v|_{H^s(D)} . \tag{18.6}$$

The related Deny–Lions lemma states that the seminorm $|\cdot|_{H^s}$ is a norm on the quotient $H^s(D)/\mathbb{P}_{s-1}(D)$, equivalent there to the quotient norm. This explains why the right-hand side in (18.6) contains a seminorm: the polynomial part is reproduced exactly and should not contribute to the error.

18.3 The local Lagrange interpolant

For a sufficiently regular function v, define $I_T v \in \mathbb{P}_k(T)$ by

$$(I_T v)(z) = v(z) \qquad \forall z \in \mathcal{N}_k(T). \tag{18.7}$$

Point values are continuous functionals on $H^s(T)$ only when

$$s > \frac{d}{2}, \tag{18.8}$$

by the Sobolev embedding theorem. Under this condition, I_T is a bounded operator on the reference element and reproduces $\mathbb{P}_k$. Bramble–Hilbert and affine scaling give the fundamental local estimate.

Theorem 18.3 (Local Lagrange interpolation)**.** *Let $\{\mathcal{T}_h\}_h$ be shape regular, let $0 \le m \le s \le k+1$ be integers, and suppose $s > d/2$. Then*

$$\|v - I_T v\|_{H^m(T)} \le C h_T^{s-m} \, |v|_{H^s(T)} \qquad \forall v \in H^s(T), \tag{18.9}$$

where C is independent of T and h_T.

For the most frequently used cases $m = 0, 1$, this reads

$$\|v - I_T v\|_{L^2(T)} \le C h_T^{s} \, |v|_{H^s(T)} \,, \tag{18.10}$$

$$|v - I_T v|_{H^1(T)} \le C h_T^{s-1} \, |v|_{H^s(T)} \,. \tag{18.11}$$

Nodal interpolation is not defined on all of H^1

In dimension two or three, an H^1 function need not have meaningful point values. Thus the nodal operator I_h cannot be applied to an arbitrary energy-space function. For low-regularity approximation one uses a quasi-interpolant such as the Clément or Scott–Zhang operator, whose degrees of freedom are local averages rather than point values.

18.4 Global interpolation estimates

The local interpolants agree on common faces because they interpolate the same trace at the shared Lagrange nodes. They therefore define a global function $I_h v \in V_h^k$. Squaring the local estimates and summing over the elements yields

$$\sum_{T \in \mathcal{T}_h} \|v - I_h v\|^2_{H^m(T)} \le C \sum_{T \in \mathcal{T}_h} h_T^{2(s-m)} \, |v|^2_{H^s(T)} \,. \tag{18.12}$$

If $v \in H^s(\Omega)$ and $m = 0$ or 1, then

$$\|v - I_h v\|_{H^m(\Omega)} \le C h^{s-m} \, \|v\|_{H^s(\Omega)} \,. \tag{18.13}$$

The restriction to $m \le 1$ is natural for a C^0 Lagrange function: globally it lies in H^1, but generally not in H^2, even though its restriction to each element is smooth.

On a nonuniform mesh, the sharper elementwise form (18.12) should be retained. It shows where refinement is useful and is the starting point for adaptive methods.

For a quasi-interpolant $\Pi_h : H^1(\Omega) \to V_h^k$, one has patchwise estimates of the form

$$\|v - \Pi_h v\|_{H^m(T)} \le C h_T^{s-m} \, |v|_{H^s(\omega_T)} , \qquad m = 0, 1, \tag{18.14}$$

where ω_T is the union of elements sharing a node with T. Finite overlap of these patches yields the corresponding global bound.

18.5 Finite element error estimates

Consider a coercive second-order elliptic problem with energy space $V = H_0^1(\Omega)$ and conforming space $V_{h,0}^k$. Combining Céa's lemma with (18.13) gives the basic a priori estimate.

Theorem 18.4 (Energy-norm convergence)**.** *Let the mesh family be shape regular and suppose $u \in H^s(\Omega) \cap H_0^1(\Omega)$, with $1 < s \le k+1$ and $s > d/2$ when the nodal interpolant is used. Then*

$$\|u - u_h\|_{H^1(\Omega)} \le C h^{s-1} \, \|u\|_{H^s(\Omega)} . \tag{18.15}$$

In particular, if $u \in H^{k+1}(\Omega)$, then $\|u - u_h\|_{H^1} = \mathcal{O}(h^k)$.

For a solution having only H^{1+r}-regularity, with $0 < r \le k$, the estimate becomes

$$\|u - u_h\|_{H^1(\Omega)} \le C h^r \, \|u\|_{H^{1+r}(\Omega)} . \tag{18.16}$$

A quasi-interpolant supplies this result even when nodal values are not defined.

Suppose next that the adjoint problem with right-hand side $g \in L^2(\Omega)$ has the regularity estimate

$$\|z\|_{H^{1+\alpha}(\Omega)} \le C_{\mathrm{reg}} \, \|g\|_{L^2(\Omega)} , \qquad 0 < \alpha \le 1. \tag{18.17}$$

Repeating the Aubin–Nitsche argument gives

$$\|u - u_h\|_{L^2(\Omega)} \le C h^{\alpha} \, \|u - u_h\|_{H^1(\Omega)} . \tag{18.18}$$

Thus full H^2 adjoint regularity, $\alpha = 1$, supplies one additional power of h; on a nonconvex polygon, α may be smaller.

For smooth solutions on regular meshes, further estimates are possible: maximum-norm bounds describe pointwise error, and special symmetry may produce *superconvergence* at selected points or for recovered gradients. These are not automatic consequences of Céa's lemma and are more sensitive to dimension, polynomial degree, mesh geometry, and logarithmic factors.

The finite element error pipeline

regularity of u $\longrightarrow$ local polynomial approximation

Galerkin error $\longleftarrow$ global best approximation

An L^2-estimate appends a fifth box: regularity of the adjoint problem. Every convergence rate can be traced to one of these inputs.

Exercises

Exercise 18.1 (Scaling of Sobolev seminorms)**.** In one space dimension, map $(0,1)$ to $(0,h)$ and derive the exact scaling of the L^2-norm and the H^m-seminorm. Compare the powers of h with (18.1).

Exercise 18.2 (Why shape regularity matters)**.** Consider triangles with vertices $(0,0)$, $(1,0)$, and $(0,\varepsilon)$. Compute the norms of the affine map and its inverse as $\varepsilon \to 0$, and explain why a uniform interpolation constant cannot follow from diameter control alone.

Exercise 18.3 (A local-to-global estimate)**.** Starting from (18.11), square and sum over the elements to prove (18.13) for $m = 1$.

Exercise 18.4 (Partial adjoint regularity)**.** Assume (18.17). Rework the duality proof to establish (18.18); state the approximation estimate needed for the dual solution.

19 Conditioning and the solution of finite element systems

Approximation theory explains how the discretization error decreases as the mesh is refined. A second question is how difficult it is to solve the resulting linear system. These issues are linked but distinct: refinement improves approximation while usually making the unpreconditioned algebraic system more ill-conditioned.

19.1 Spectral condition numbers

For an invertible matrix $\mathbf{A}$, the condition number in a chosen matrix norm is

$$\operatorname{cond}(\mathbf{A}) = \|\mathbf{A}\| \left\|\mathbf{A}^{-1}\right\|. \tag{19.1}$$

If $\mathbf{A}$ is symmetric positive definite and the Euclidean norm is used, then

$$\operatorname{cond}_2(\mathbf{A}) = \frac{\lambda_{\max}(\mathbf{A})}{\lambda_{\min}(\mathbf{A})}. \tag{19.2}$$

The condition number measures worst-case amplification of relative perturbations and strongly influences convergence of many iterative methods. It is a property of the matrix representation, so it depends on the chosen basis as well as on the underlying variational problem.

Let $\mathbf{A}$ be the matrix obtained after eliminating Dirichlet degrees of freedom for a coercive, symmetric second-order elliptic problem, and let

$$v_h = \sum_{i=1}^{n} V_i \phi_i. \tag{19.3}$$

Then

$$\mathbf{V}^\top \mathbf{A} \mathbf{V} = a(v_h, v_h). \tag{19.4}$$

The Rayleigh quotient translates spectral bounds into finite element norm estimates.

19.2 Coefficient norms and the inverse inequality

On a quasi-uniform, shape-regular mesh with the standard fixed-degree nodal basis, the coefficient vector and the L^2-norm are related by

$$c h^d \, |\mathbf{V}|_2^2 \le \|v_h\|_{L^2(\Omega)}^2 \le C h^d \, |\mathbf{V}|_2^2 . \tag{19.5}$$

The factor h^d is the volume of a typical basis-function support. The uniform constants follow from reference-element scaling and the fact that only a bounded number of elements meet at each node.

The second ingredient reverses the usual direction of a Sobolev estimate.

Lemma 19.1 (Finite element inverse inequality)**.** *Let V_h^k have fixed degree k on a shape-regular mesh. Then*

$$|v_h|_{H^1(T)} \le C h_T^{-1} \, \|v_h\|_{L^2(T)} \qquad \forall v_h \in V_h^k, \tag{19.6}$$

and, on a quasi-uniform mesh,

$$|v_h|_{H^1(\Omega)} \le C h^{-1} \, \|v_h\|_{L^2(\Omega)} . \tag{19.7}$$

On the finite-dimensional reference polynomial space, all norms are equivalent. Applying this fact to $\widehat{v}_h$ and scaling to T proves (19.6). Unlike the interpolation estimate, this inequality applies only to discrete functions. It says that a degree-k polynomial on a cell cannot oscillate on a length scale smaller than the cell itself.

19.3 Conditioning of the stiffness matrix

Assume that

$$c_a \, |v|^2_{H^1(\Omega)} \le a(v,v) \le C_a \, \|v\|^2_{H^1(\Omega)} \qquad \forall v \in H^1_0(\Omega). \tag{19.8}$$

The lower spectral bound follows from coercivity, Poincaré's inequality, and the left inequality in (19.5):

$$\mathbf{V}^\top \mathbf{A} \mathbf{V} \ge c \, \|v_h\|^2_{H^1(\Omega)} \ge c \, \|v_h\|^2_{L^2(\Omega)} \ge c h^d \, |\mathbf{V}|_2^2 \, . \tag{19.9}$$

Thus $\lambda_{\min}(\mathbf{A}) \ge c h^d$. For the upper bound, use the inverse inequality and the other half of the coefficient equivalence:

$$\mathbf{V}^\top \mathbf{A} \mathbf{V} \le C \, \|v_h\|^2_{H^1(\Omega)} \le C h^{-2} \, \|v_h\|^2_{L^2(\Omega)} \le C h^{d-2} \, |\mathbf{V}|_2^2 \, . \tag{19.10}$$

Hence $\lambda_{\max}(\mathbf{A}) \le C h^{d-2}$, and we obtain the standard estimate.

Theorem 19.2 (Stiffness-matrix condition number)**.** *For a fixed-degree conforming discretization of a uniformly elliptic second-order problem on a quasi-uniform, shape-regular mesh,*

$$\operatorname{cond}_2(\mathbf{A}) \le C h^{-2}. \tag{19.11}$$

The estimate concerns the symmetric positive definite matrix after the Dirichlet degrees of freedom have been eliminated, or after a symmetry-preserving equivalent modification.

The power h^{-2} is independent of the space dimension, although the number of unknowns is not. On a quasi-uniform mesh, $n \simeq h^{-d}$, so

$$\operatorname{cond}_2(\mathbf{A}) = \mathcal{O}(n^{2/d}). \tag{19.12}$$

For locally refined meshes the smallest elements can dominate the largest eigenvalue, and a single global h no longer describes the spectrum sharply.

19.4 Direct and iterative solvers

A sparse direct method computes a factorization such as $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$. It is robust and is attractive for moderate problem sizes or for many right-hand sides, but elimination creates *fill-in*. Representative complexities for well-ordered elliptic grid problems are

	one dimension	two dimensions	three dimensions
factorization work	$\mathcal{O}(n)$	$\mathcal{O}(n^{3/2})$	$\mathcal{O}(n^2)$
factor storage	$\mathcal{O}(n)$	$\mathcal{O}(n \log n)$	$\mathcal{O}(n^{4/3})$

These estimates assume banded elimination in one dimension and good nested-dissection-type orderings in higher dimensions; poor ordering can be substantially more expensive.

Stationary iterations split $\mathbf{A} = \mathbf{M} - \mathbf{N}$ and update

$$\mathbf{U}^{(m+1)} = \mathbf{M}^{-1}\mathbf{N}\mathbf{U}^{(m)} + \mathbf{M}^{-1}\mathbf{F}. \tag{19.13}$$

Jacobi and Gauss–Seidel are simple examples. They rapidly reduce oscillatory error components but are slow on smooth error components as the mesh is refined. This smoothing property, apparently a weakness as a stand-alone solver, is exactly what multigrid exploits.

For a symmetric positive definite matrix, the conjugate-gradient method constructs iterates in the Krylov spaces

$$\mathbf{U}^{(0)} + \mathcal{K}_m(\mathbf{A}, \mathbf{r}^{(0)}), \qquad \mathcal{K}_m(\mathbf{A}, \mathbf{r}) = \operatorname{span}\left\{\mathbf{r}, \mathbf{A}\mathbf{r}, \dots, \mathbf{A}^{m-1}\mathbf{r}\right\}, \tag{19.14}$$

where $\mathbf{r}^{(0)} = \mathbf{F} - \mathbf{A}\mathbf{U}^{(0)}$. It minimizes the energy norm of the algebraic error over this affine space and satisfies

$$\left\|\mathbf{e}^{(m)}\right\|_{\mathbf{A}} \le 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{m}\left\|\mathbf{e}^{(0)}\right\|_{\mathbf{A}}, \qquad \kappa = \operatorname{cond}_2(\mathbf{A}). \tag{19.15}$$

Thus the pessimistic iteration count grows like $\mathcal{O}(\sqrt{\kappa})$. With $\kappa = \mathcal{O}(h^{-2})$, unpreconditioned CG may require $\mathcal{O}(h^{-1})$ iterations.

For nonsymmetric matrices, methods such as GMRES replace CG; for symmetric indefinite matrices, MINRES is a natural choice. The matrix structure, not merely sparsity, determines which Krylov method is appropriate.

19.5 Preconditioning

A preconditioner $\mathbf{P}$ is an easily invertible approximation to $\mathbf{A}$. One solves an equivalent system such as

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{U} = \mathbf{P}^{-1}\mathbf{F}. \tag{19.16}$$

For CG, $\mathbf{P}$ must be symmetric positive definite and the preconditioned algorithm is interpreted in the corresponding inner product. The objective is not necessarily to approximate every entry of $\mathbf{A}$, but to make the spectra of $\mathbf{P}$ and $\mathbf{A}$ equivalent:

$$c\,\mathbf{V}^{\top}\mathbf{P}\mathbf{V} \le \mathbf{V}^{\top}\mathbf{A}\mathbf{V} \le C\,\mathbf{V}^{\top}\mathbf{P}\mathbf{V} \qquad \forall \mathbf{V}, \tag{19.17}$$

with c, C independent of h. Then the relevant condition number is bounded independently of mesh refinement.

Diagonal scaling, incomplete factorizations, overlapping or nonoverlapping domain decomposition, and multigrid are common choices. Their tradeoffs include setup cost, memory, parallelism, and robustness to coefficient variation and anisotropy.

19.6 Geometric multigrid

Suppose that the meshes are nested and give spaces

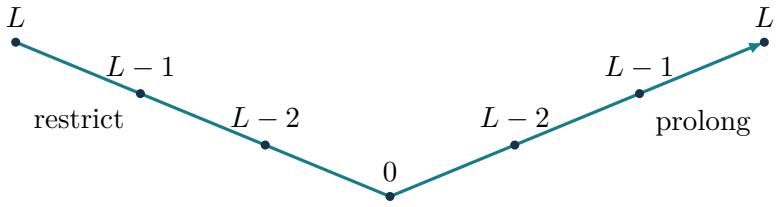
$$V_0 \subset V_1 \subset \cdots \subset V_L, \tag{19.18}$$

with stiffness matrices $\mathbf{A}_\ell$. A smoother efficiently removes the high-frequency part of the error on level ℓ; the remaining smooth error can be represented on the coarser space $V_{\ell-1}$. Let $\mathbf{I}_{\ell-1}^{\ell}$ denote prolongation and $\mathbf{I}_{\ell}^{\ell-1}$ restriction, often chosen as its transpose.

One V-cycle for $\mathbf{A}_\ell\mathbf{u}_\ell = \mathbf{f}_\ell$ is:

1. apply ν_1 pre-smoothing steps to $\mathbf{u}_\ell$;
2. form the residual $\mathbf{r}_\ell = \mathbf{f}_\ell - \mathbf{A}_\ell\mathbf{u}_\ell$;
3. restrict it: $\mathbf{r}_{\ell-1} = \mathbf{I}_{\ell}^{\ell-1}\mathbf{r}_\ell$;

4. approximately solve $\mathbf{A}_{\ell-1}\mathbf{e}_{\ell-1} = \mathbf{r}_{\ell-1}$ by one or more recursive cycles, solving exactly on level zero;
5. prolong and correct:
$$\mathbf{u}_\ell \leftarrow \mathbf{u}_\ell + \mathbf{I}_{\ell-1}^{\ell}\mathbf{e}_{\ell-1}; \tag{19.19}$$
6. apply ν_2 post-smoothing steps.



Under standard approximation and smoothing assumptions, a multigrid cycle reduces the error by a factor bounded away from one independently of h, while its work is proportional to the number of unknowns. It can be used as a solver or as a preconditioner for CG.

19.7 Discretization error versus algebraic error

Let u_h be the exact Galerkin solution and let $\widetilde{u}_h$ be the function represented by an inexact algebraic iterate. The total error satisfies

$$\|u - \widetilde{u}_h\|_V \le \underbrace{\|u - u_h\|_V}_{\text{discretization error}} + \underbrace{\|u_h - \widetilde{u}_h\|_V}_{\text{algebraic error}}. \tag{19.20}$$

There is little benefit in reducing the algebraic error many orders of magnitude below the discretization error. A practical stopping criterion should balance the two, preferably using a residual norm related to the energy norm through a robust preconditioner.

Mesh refinement creates two simultaneous trends

For degree k and a smooth solution, the energy error decreases like h^k, while the condition number of the unpreconditioned stiffness matrix grows like h^{-2}. Optimal approximation therefore needs an optimal solver: preconditioning and multilevel methods are part of the numerical method, not an afterthought.

Exercises

Exercise 19.1 (Condition number in terms of unknowns)**.** On a quasi-uniform mesh in dimension d, use $n \simeq h^{-d}$ to derive (19.12). Compare the result in dimensions one, two, and three.

Exercise 19.2 (A Rayleigh-quotient proof)**.** Starting from (19.4), prove (19.9) and (19.10), listing every assumption used at each step.

Exercise 19.3 (CG without preconditioning)**.** Combine (19.15) with (19.11). Estimate the number of iterations needed to reduce the energy error by a fixed factor as $h \to 0$, and combine this with the cost of one sparse matrix–vector product.

Exercise 19.4 (Dirichlet enforcement)**.** For a symmetric 3×3 matrix, prescribe the third component of the solution. Carry out block elimination and compare it with (i) clearing both the third row and column and (ii) replacing only the third row. Which constructions preserve symmetry?

Exercise 19.5 (One two-grid correction)**.** Explain why the coarse-grid equation uses the restricted residual rather than the restricted current iterate. Show that if an error lies exactly in the range of the prolongation operator and the coarse solve is exact, the coarse-grid correction removes it.

20 Notation at a glance

Symbol	Meaning
$I_h u$	restriction or interpolation of u to a discrete space
δ_h^2	centered second difference on a uniform grid
$\mathcal{L}_h$	discrete differential operator
τ_h	truncation error or discrete residual of the exact solution
$\|\cdot\|_{\ell^\infty}$	maximum over the relevant grid nodes
H_0^1	H^1 functions with zero trace
H^{-1}	dual of H_0^1
V_h	conforming finite element subspace
$a(\cdot,\cdot)$	continuous bilinear form
A_h	algebraic matrix of a discrete problem

Suggested reading

For elliptic PDE theory and maximum principles, distributions, Sobolev spaces, and elliptic regularity, see [2, 1]. Weak formulations and finite elements are developed in [3, 4, 5, 6, 7]. Broad treatments of finite differences, boundary closures, and numerical PDEs appear in [7, 8, 9].

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